NeoSignal

AI Intelligence Daily · Multi-Source · Cross-Verified

11 August 2026

255 verified · 508 raw · 5 sources

255	5	250	0
TOTAL	VERIFIED	CONFIRMED	EMERGING

Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

VERIFIED	**0.80 - 1.00**	3+ independent sources including media
CONFIRMED	**0.50 - 0.79**	2 sources or 1 high-quality media
EMERGING	**0.25 - 0.49**	Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	236	Media
TechCrunch AI	9	Media
HackerNews	8	Community
MIT Tech Review	4	Media
The Verge AI	4	Media

VERIFIED INTELLIGENCE — 5 stories

VERI

AI professors are negotiating the new realities of academic research

MIT Tech Review +2 sources **100%**

This story originally appeared in The Algorithm, our weekly newsletter on AI. To get stories like this in your inbox first, sign up here. Last week, I headed 30 miles south of San Francisco to a hotel in Mountain View, California, to join some of the most accomplished, and some...

https://www.technologyreview.com/2026/08/10/1141597/ai-professors-are-negotiating-the-n...

VERI

Intelligence Foundation Model: A New Perspective to Approach Artificial General Intelligence

ArXiv CS.AI +1 source **80%**

arXiv:2511.10119v4 Announce Type: replace Abstract: We propose a new perspective for approaching artificial general intelligence (AGI) through an intelligence foundation model (IFM). Unlike existing foundation models (FMs), which specialize in pattern learning within specific...

https://arxiv.org/abs/2511.10119

VERI

SHE: Trajectory-driven Safety Harness Evolution for LLM Agents

ArXiv CS.AI +1 source **80%**

arXiv:2608.09885v1 Announce Type: new Abstract: The safety of large language model (LLM) agents depends not only on model weights but also on the agent harness that manages context, memory, tools, permissions, and runtime control. Existing safety mechanisms often treat the...

https://arxiv.org/abs/2608.09885

VERI

UniSpace: Unified Visual Representation and Scalable Multimodal Modeling

ArXiv CS.AI +1 source **80%**

arXiv:2608.08676v1 Announce Type: cross Abstract: Semantic vision encoders have become a central visual interface for multimodal understanding and semantic conditioning in image generation. However, their final tokens discard fine-grained visual details, leading to poor pixel...

https://arxiv.org/abs/2608.08676

VERI

TransNRank: Towards Accurate Neoantigen Ranking with Transformer

ArXiv CS.AI +1 source **80%**

arXiv:2608.01924v2 Announce Type: replace-cross Abstract: Personalized neoantigen prediction is challenging due to the scarcity of positive samples, the noise of the experimental data, the severe class imbalance trait and the complex of immunogenicity features. Prior arts, such...

https://arxiv.org/abs/2608.01924

CONFIRMED SIGNALS — 250 stories

CONF

Show HN: Needle2: 14MB agentic LLM for phones, wearables, smart home and robots

HackerNews **60%** HN 238

Hey HN, Henry from Cactus here! We previously released Cactus Needle, a 14MB agentic LLM for tool call, device use, and structured extraction for phones, wearables, smart homes, small robots and microcontrollers. We got really great feedback here, and have now incorporated the...

https://cactuscompute.com/needle

CONF

Show HN: A tiny LLM running at 21,000 tok/s on a $250 FPGA (Live Demo)

HackerNews **50%** HN 46

No summary — see link for full article.

https://www.mikeayles.com/blog/on-chip-llm-kv260/

CONF

Show HN: Neolabs.fyi – 100 new AI labs by research area, valuation, and more

HackerNews **50%** HN 8

No summary — see link for full article.

https://neolabs.fyi/

CONF

Show HN: Pyrig – A tool that automates project setup and maintenance

HackerNews **50%** HN 4

During the implementation of a huge high performance service. In order to keep context small (mainly for humans) I kept the specs into mermaid diagrams. When communicating with humans; diagrams were easy to follow and to remember. But when I asked the agent to implement what's...

https://github.com/Winipedia/pyrig

0 **CONF**

Show HN: PrivateRedact – Offline PII redaction with a local LLM, no cloud

HackerNews **50%** HN 4

Hi HN, I've been building OptiQra for quite a while, and I finally finished the desktop version. OptiQra is an open-source website intelligence and optimization platform. It can crawl and analyze websites or local projects, identify SEO/AEO/GEO, accessibility, performance and...

https://github.com/monjurulkarim/privateredact

1 **CONF**

Show HN: TickClip – An AI shopping agent that can recommend not buying

HackerNews **50%** HN 2

My name is Tarik Aarbaoui, I am an affiliate and influencer marketing specialist. For the last 14 years I’ve worked to grow revenue while trying to meet shoppers’ expectations. But during my time there, I observed a core conflict: shoppers often look to influencers and affiliate...

https://www.tickclip.ai/

2 **CONF**

Show HN: Mangudai – Agentless attack surface manager for small tech teams

HackerNews **50%** HN 2

No summary — see link for full article.

https://github.com/oktaybilge1/mangudai-overview

3 **CONF**

Show HN: An open-source multi-tenant, AI-native software factory

HackerNews **50%** HN 2

No summary — see link for full article.

https://github.com/missingstudio/eva

4 **CONF**

OpenAI reportedly completed a $7 billion employee tender offer

TechCrunch AI **50%**

San Francisco's housing market is in trouble again.

https://techcrunch.com/2026/08/10/openai-reportedly-completed-a-7-billion-employee-tend...

5 **CONF**

As AI-led attacks multiply, OpenAI launches a new cyber model

TechCrunch AI **50%**

OpenAI is expanding its AI cybersecurity defense program Daybreak, and rolling out a new cyber-trained AI model with it.

https://techcrunch.com/2026/08/10/as-ai-led-attacks-multiply-openai-launches-a-new-cybe...

6 **CONF**

Mark Zuckerberg’s AI manifesto is exactly why people don’t like AI

TechCrunch AI **50%**

On Monday, Mark Zuckerberg published a 6,500-word manifesto about personal AI, largely about the possibilities for the "personal superintelligence" systems Meta AI is building.

https://techcrunch.com/2026/08/10/mark-zuckerbergs-ai-manifesto-is-exactly-why-people-d...

7 **CONF**

Tech industry is buzzing after a Claude agent hacked into a gym

TechCrunch AI **50%**

An OpenClaw agent hacked into a gym's reservation system to bump its human boss higher on a class' waitlist. And the tech industry took notice.

https://techcrunch.com/2026/08/10/tech-industry-is-buzzing-after-a-claude-agent-hacked-...

8 **CONF**

Discovered Materials is playing AI whack-a-mole to hunt cooler chips

TechCrunch AI **50%**

Discovered Materials raised $9 million to fund the hunt for more novel materials to build more efficient chips.

https://techcrunch.com/2026/08/10/discovered-materials-is-playing-ai-whack-a-mole-to-hu...

9 **CONF**

ChatGPT brings unlimited text chats to free users

TechCrunch AI **50%**

OpenAI said that ChatGPT free and Go users are also getting a new think button for complex queries.

https://techcrunch.com/2026/08/06/openai-brings-unlimited-chatgpt-text-chats-to-free-us...

0 **CONF**

The Download: AI agents for science, and the “censorship-industrial complex”

MIT Tech Review **50%**

This is today’s edition of The Download, our weekday newsletter that provides a daily dose of what’s going on in the world of technology. AI for science needs reasoning, not just data —Eric Schmidt, the former CEO of Google and the cofounder of Schmidt Sciences, and Suhas...

https://www.technologyreview.com/2026/08/10/1141526/the-download-ai-agents-science-cens...

1 **CONF**

The Download: a censorship conspiracy theory and the first virus created by AI

MIT Tech Review **50%**

This is today’s edition of The Download, our weekday newsletter that provides a daily dose of what’s going on in the world of technology. How ideas of a vast censorship network moved from the online fringe to Trump policy For years, narratives about a “censorship-industrial...

https://www.technologyreview.com/2026/08/07/1141389/the-download-censorship-conspiracy-...

2 **CONF**

Four takeaways from Mark Zuckerberg’s massive AI manifesto

The Verge AI **50%**

Meta CEO Mark Zuckerberg has a lot to say about the idealized future he now envisions for humanity co-existing with artificial intelligence - his latest essay spans more than 6,500 words on the matter. The lengthy manifesto Zuckerberg published on Monday, titled "The Future is...

https://www.theverge.com/tech/977395/meta-mark-zuckerberg-superintelligent-ai-ramble

3 CONF

What happens to Bose when headphones become AI?

The Verge AI 50%

Today, I’m talking with Lila Snyder, who is the CEO of Bose. You certainly know Bose — it’s one of the most famous brands in all of consumer tech. The company started 60 years ago selling speakers to consumers, and its focus on research and development has led it to be a leader...

https://www.theverge.com/podcast/975732/bose-ceo-lila-snyder-ai-wearables-licensing-hea...

4 CONF

Ford’s new AI assistant can check your fuel levels and tire pressure

The Verge AI 50%

Ford is rolling out a new AI-powered assistant that can answer questions about your Ford or Lincoln vehicle, such as how much fuel you'll need for your next road trip or whether your truck can tow that new motor boat. The app is rolling out first to the Ford and Lincoln mobile...

https://www.theverge.com/transportation/976748/ford-ai-assistant-mobile-app

5 CONF

Flow-by-Flow:Content-Judgment Bypass for Governing AI Output in High-Loss Domains

ArXiv CS.AI 50%

arXiv:2608.07474v1 Announce Type: new Abstract: Prior work showed that human-in-the-loop oversight becomes structurally untenable in high-loss domains when AI output velocity V exceeds human cognitive capacity C_max. The operative constraint, however, is not V alone but V x L,...

https://arxiv.org/abs/2608.07474

6 CONF

Training Variable Long Sequences with Data-Centric Parallel

ArXiv CS.AI 50%

arXiv:2608.07524v1 Announce Type: new Abstract: Training deep learning models on variable long sequences poses significant computational challenges. Existing methods force a difficult trade-off between efficiency and ease-of-use. Simple approaches use static configurations that...

https://arxiv.org/abs/2608.07524

7 CONF

CORDA: A Benchmark for Hierarchical Harm-Centric Moral Reasoning in Large Language Models

ArXiv CS.AI 50%

arXiv:2608.08061v1 Announce Type: new Abstract: The key question in moral judgement is not simply whether someone chooses the "right" answer, but how they decide what matters most when moral principles conflict. Current evaluations of large language models (LLMs) remain limited:...

https://arxiv.org/abs/2608.08061

8 CONF

Bidirectional Context Self-Distillation for Reinforcement Learning of Skill-Based LLM Agents

ArXiv CS.AI 50%

arXiv:2608.09555v1 Announce Type: new Abstract: External natural-language skills provide large language model (LLM) agents with reusable and editable guidance for solving complex tasks. Yet their effectiveness depends not only on skill quality, but also on whether the policy can...

https://arxiv.org/abs/2608.09555

9 **CONF**

When LLM Agents Negotiate: Private Information and Dynamic Bargaining in Supply Chains

ArXiv CS.AI 50%

arXiv:2608.07538v1 Announce Type: new Abstract: As LLM agents move from decision support to autonomous procurement, firms need to know whether delegated negotiators create value, divide it predictably, and avoid money-losing contracts. We study this in a canonical supply chain...

https://arxiv.org/abs/2608.07538

0 **CONF**

An AI Scientist that Doesn't Drift: Taste, Structure, and Falsifiable Findings in a Quadruped Navigation Research Loop

ArXiv CS.AI 50%

arXiv:2608.07542v1 Announce Type: new Abstract: Autonomous research loops driven by large language models can run machine-learning experiments at scale but tend to drift toward local refinements of whichever metric they optimise rather than testing the hypotheses that motivate...

https://arxiv.org/abs/2608.07542

1 **CONF**

TeXFix-Bench: An Empirically Grounded Multi-Format Benchmark for LLM-Based Document Source Repair

ArXiv CS.AI 50%

arXiv:2608.07617v1 Announce Type: new Abstract: Scientific and technical writing depends on markup sources that must compile: LaTeX, Typst, and Markdown pipelines fail on missing delimiters, mismatched environments, broken imports, or package conflicts. Existing document-repair...

https://arxiv.org/abs/2608.07617

2 **CONF**

CMU-Drive and V2V-VLA: Cooperative Multi-agent Unified Driving with Reasoning Benchmark and Vehicle-to-Vehicle Vision-Language-Action Models

ArXiv CS.AI 50%

arXiv:2608.07621v1 Announce Type: new Abstract: Vision-Language-Action (VLA) models have recently achieved impressive performance for end-to-end autonomous driving, yet existing approaches are primarily designed for an individual single autonomous driving agent with limited...

https://arxiv.org/abs/2608.07621

3 **CONF**

Improving Constraint Models with LLM Agents

ArXiv CS.AI 50%

arXiv:2608.08127v1 Announce Type: new Abstract: The runtime of Constraint Programming (CP) solvers is highly sensitive to modeling choices, such as symmetry breaking, implied constraints, global constraints, constraint reformulation, and variable representation. Improving these...

https://arxiv.org/abs/2608.08127

4 **CONF**

From Single Chatbots to Governed Agent Ecosystems: An Agentic AI Pattern Catalogue and Orchestration Framework for Mission-Critical Hospital Information Management Systems

ArXiv CS.AI 50%

arXiv:2608.07627v1 Announce Type: new Abstract: Hospitals are racing to embed AI, while coping with the surge in adaptation of the technology in other industries, into the triage management, documentation, scheduling, and revenue-cycle workflows, yet most deployments remain as...

https://arxiv.org/abs/2608.07627

5 CONF

Agent-MD: Selective LLM Intervention with Event-Driven Escalation for Stateful GCMC--MD Campaigns

ArXiv CS.AI 50%

arXiv:2608.07637v1 Announce Type: new Abstract: Long-running molecular simulation campaigns require repeated continuation from saved states, provenance-aware progression, adaptive assessment, and occasional interpretation of workflow conditions that cannot be resolved safely by...

https://arxiv.org/abs/2608.07637

6 CONF

Contextual Value Alignment via Multilayer Combinatorial Fusion

ArXiv CS.AI 50%

arXiv:2608.07642v1 Announce Type: new Abstract: Aligning large language models (LLMs) with human values remains a major challenge, especially for trustworthy AI. While existing approaches such as RLHF, CAI, and their variants have achieved promising results, they often rely on a...

https://arxiv.org/abs/2608.07642

7 CONF

Decoding Phenotypes: A Framework for Fusing Genomic Language Models and Neuroimaging

ArXiv CS.AI 50%

arXiv:2608.08926v1 Announce Type: new Abstract: Neuroimaging and genetic testing are two important clinical references for nervous system diseases, offering complementary diagnostic information. However, integrating genomic and neuroimaging data for precise disease diagnosis is...

https://arxiv.org/abs/2608.08926

8 CONF

IntelliAudit: Using Large Language Models to Evaluate Audit Controls

ArXiv CS.AI 50%

arXiv:2608.07688v1 Announce Type: new Abstract: IT audits require auditors to judge whether heterogeneous organizational evidence satisfies semantic security and compliance controls. This judgment is difficult to automate because relevant evidence is distributed across policies,...

https://arxiv.org/abs/2608.07688

9 CONF

Protecting patient privacy in clinical foundation models: Technical and legal perspectives

ArXiv CS.AI 50%

arXiv:2608.07705v1 Announce Type: new Abstract: Clinical foundation models trained on large-scale patient data are increasingly used for decision support, screening, and public health. As deployment expands, privacy risk increasingly arises from model-mediated leakage, yet its...

https://arxiv.org/abs/2608.07705

0 CONF

EMMR: Emotion-Mediated Multimodal Reasoning for Personality Assessment in Asynchronous Video Interviews

ArXiv CS.AI 50%

arXiv:2608.07512v1 Announce Type: cross Abstract: Asynchronous Video Interviews (AVIs) have become increasingly popular for personality assessment. Recent large language models (LLMs) have shown potential for personality assessment from transcribed interview responses. However,...

https://arxiv.org/abs/2608.07512

1 CONF

The Capability Ladder: A Curriculum-Modernization Framework for Workforce Readiness in the AI Era

ArXiv CS.AI 50%

arXiv:2608.07779v1 Announce Type: new Abstract: Artificial intelligence is changing the task composition of computing work faster than curricula and training typically adapt. This is a curriculum-framework paper, grounded in a structured narrative review of labor-market and...

https://arxiv.org/abs/2608.07779

2 CONF

Who Built This Model? Tracing LLM Lineage via Spectral Fingerprints in Weight Space

ArXiv CS.AI 50%

arXiv:2608.07786v1 Announce Type: new Abstract: Open-weight large language models (LLMs) are increasingly developed through complex, multi-stage pipelines, leading to intricate lineage relationships that reflect model origin, ownership, and evolution. Understanding these...

https://arxiv.org/abs/2608.07786

3 CONF

When the Judge Should Not Decide: Evidence-Locked, Non-Compensatory Selection Bounds LLM-Judge Failure in Reasoning Pipelines

ArXiv CS.AI 50%

arXiv:2608.07813v1 Announce Type: new Abstract: An LLM judge deployed inside a reasoning pipeline does not merely measure quality, it decides which answer ships. We show that the cost of that decision depends less on judge accuracy than on the decision rule the judge is embedded...

https://arxiv.org/abs/2608.07813

4 CONF

Counterfactual Benchmarking and Training for Factuality Consistency and Order-Robust Grounded Reasoning in LLMs over Heterogeneous Knowledge

ArXiv CS.AI 50%

arXiv:2608.07838v1 Announce Type: new Abstract: Large language models (LLMs) have increasingly supported response generation grounded in user-provided knowledge spanning heterogeneous structures. However, existing benchmarks provide limited assessment of whether LLMs can...

https://arxiv.org/abs/2608.07838

5 CONF

Back to the Future: A workbook time machine for spread sheet creation benchmarks

ArXiv CS.AI 50%

arXiv:2608.07873v1 Announce Type: new Abstract: We introduce the workbook time machine, a pipeline that automatically creates benchmarks evaluating the ability of language models to create derived objects in spreadsheets (formulas, charts, pivot tables, and conditional...

https://arxiv.org/abs/2608.07873

6 CONF

GRACE: LLM-Grounded Semantic Metric Spaces for Scalable Mixed-Data Clustering

ArXiv CS.AI 50%

arXiv:2608.07881v1 Announce Type: new Abstract: Clustering mixed tabular data requires a unified metric space to bridge the inherent heterogeneity between continuous numerical measurements and discrete categorical symbols. Traditionally, algorithms rely entirely on...

https://arxiv.org/abs/2608.07881

7 CONF

TelemetrySuffBench: Is Agent Telemetry Sufficient for Failure-Origin Diagnosis?

ArXiv CS.AI 50%

arXiv:2608.07899v1 Announce Type: new Abstract: Agent systems increasingly expose execution traces, yet telemetry that reveals a failure may still be inadequate for identifying where that failure originated. We introduce TelemetrySuffBench, a controlled benchmark that separates...

https://arxiv.org/abs/2608.07899

8 CONF

GraphThink: Graph-Enhanced LLM Thinking for Long-Horizon Embodied Task Planning

ArXiv CS.AI 50%

arXiv:2608.07905v1 Announce Type: new Abstract: Embodied agents using LLM-based planners often struggle with physical hallucinations, poor generalization to long-horizon tasks, and lack of environmental awareness. We propose GraphThink, a novel framework that integrates a task...

https://arxiv.org/abs/2608.07905

9 CONF

ComboShoppingBench: Evaluating LLM Agents for Budget-Constrained Basket Shopping with Coupons

ArXiv CS.AI 50%

arXiv:2608.09282v1 Announce Type: new Abstract: Real-world shopping often requires constructing a basket of complementary items rather than retrieving a single product. Such combo-shopping tasks arise in device setup, meal preparation, event planning, and group takeout ordering,...

https://arxiv.org/abs/2608.09282

0 CONF

REIN: Bridging the Gap between Reasoning and Reliability via Reflection and Abstention Alignment

ArXiv CS.AI 50%

arXiv:2608.07931v1 Announce Type: new Abstract: Large reasoning models (LRMs) are prone to hallucination, which undermines their reliability and poses challenges for safe deployment. Hallucinations in LRMs arise from two distinct failure sources: reasoning hallucination, where...

https://arxiv.org/abs/2608.07931

1 CONF

Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution

ArXiv CS.AI 50%

arXiv:2608.07943v1 Announce Type: new Abstract: Multi-page visually-rich document understanding (MP-VRDU) requires managing evidence that is sparse, spread across pages, and often exceeds a model's context window. Prior work has produced competing, largely untested claims about...

https://arxiv.org/abs/2608.07943

2 CONF

RAVEN-Eval: Rubric-Guided Automatic Evaluation for AI Video Generation Models Based on LMM Preference Judgement

ArXiv CS.AI 50%

arXiv:2608.09111v1 Announce Type: new Abstract: AI video generation has advanced rapidly and entered widespread commercial use. As a result, quality differences among videos produced by state-of-the-art AI video generation models~(AIVGMs) have become increasingly difficult to...

https://arxiv.org/abs/2608.09111

3 **CONF**

Guixu: Valuation-Driven Data Discovery for Autonomous AI Agents with On-Chain Attestation

ArXiv CS.AI **50%**

arXiv:2608.07949v1 Announce Type: new Abstract: Autonomous agents increasingly rely on external data to complete downstream tasks such as model training and decision support. However, existing data discovery systems remain largely retrieval-oriented: they surface candidate...

https://arxiv.org/abs/2608.07949

4 **CONF**

KGCache: Amortized Subgraph Retrieval for KG Reasoning with LLMs

ArXiv CS.AI **50%**

arXiv:2608.07954v1 Announce Type: new Abstract: Large language models can answer knowledge-intensive questions more reliably when they are grounded with knowledge graphs, but systems such as Think-on-Graph and Reasoning-on-Graph repeatedly query the same graph neighborhoods...

https://arxiv.org/abs/2608.07954

5 **CONF**

CyberAGENTS: Structured Autonomy for Agentic Gamified Learning in Cybersecurity

ArXiv CS.AI **50%**

arXiv:2608.07965v1 Announce Type: new Abstract: Gamification is especially effective in learning domains requiring active problem-solving and iterative skill-building, such as cybersecurity education. Generative AI agents offer a path to delivering such experiences adaptively at...

https://arxiv.org/abs/2608.07965

6 **CONF**

VDGR-RAG: Vectors, Directories, Graphs, and Reflection Are All You Need for Unified Reasoning over Hierarchical Enterprise Knowledge

ArXiv CS.AI **50%**

arXiv:2608.07994v1 Announce Type: new Abstract: Retrieval-Augmented Generation (RAG) is essential for enterprise knowledge question answering (QA), particularly in domains with complex product documentation like telecommunications. However, existing RAG approaches largely...

https://arxiv.org/abs/2608.07994

7 **CONF**

Eco-SoC: A Sustainable VLSI Architecture for Energy-Proportional Artificial Intelligence

ArXiv CS.AI **50%**

arXiv:2608.08761v1 Announce Type: cross Abstract: In an era defined by escalating climate change and the pervasive deployment of edge intelligence, the environmental cost of semiconductor manufacturing and operation has reached a critical threshold. As Deep Learning (DL)...

https://arxiv.org/abs/2608.08761

8 **CONF**

JustLLMGRPO: Radiographic Control for Chest X-Ray Generation

ArXiv CS.AI **50%**

arXiv:2608.08046v1 Announce Type: new Abstract: Text-conditioned chest X-ray generation aims to synthesize realistic radiographs that faithfully depict specified findings. Existing work has primarily improved quality by updating image generators, implicitly treating prompts as...

https://arxiv.org/abs/2608.08046

9 **CONF**

H2: A Dual Hybrid Semantic Data Lake Architecture for Medical Data Harmonization with Human-In-the-Loop verified, LLM Driven Metadata Annotation System

ArXiv CS.AI **50%**

arXiv:2608.08056v1 Announce Type: new Abstract: Medical data, by its nature, exhibit a high degree of heterogeneity on multiple levels ranging from (a) different modalities like images, text and time series, (b) diverse tabular schemata introduced by institutions and (c)...

https://arxiv.org/abs/2608.08056

0 **CONF**

Generative Models: Principles, Architectures, and Applications

ArXiv CS.AI **50%**

arXiv:2608.08101v1 Announce Type: new Abstract: Generative AI has emerged as one of the most transformative forces in modern artificial intelligence, reshaping how we create, imagine, and interact with digital content. From photorealistic images to coherent text, from immersive...

https://arxiv.org/abs/2608.08101

1 **CONF**

Neurosymbolic Discovery of Algebraic Graph Constructions

ArXiv CS.AI **50%**

arXiv:2608.08118v1 Announce Type: new Abstract: There are several methods for searching for graphs with prescribed properties, such as SAT solvers and specialized generators. These methods return the result as raw data: an adjacency matrix or a string encoding. The raw data...

https://arxiv.org/abs/2608.08118

2 **CONF**

Constraining ontology mappings using metaphysical choices

ArXiv CS.AI **50%**

arXiv:2608.08122v1 Announce Type: new Abstract: In this paper we discuss the foundations behind a novel methodology for the validation of semantic mappings between different data sources based upon different foundation ontologies, where the methodology builds a framework based...

https://arxiv.org/abs/2608.08122

3 **CONF**

CoRE: Consensus Rewards via Equilibrium for Test-Time Reinforcement Learning

ArXiv CS.AI **50%**

arXiv:2608.09324v1 Announce Type: new Abstract: On unlabeled test data, reinforcement learning lacks a ground-truth reward; test-time RL methods derive one from the model's own roll-outs, rewarding those that match the majority vote over N sampled answers. That vote discards a...

https://arxiv.org/abs/2608.09324

4 **CONF**

When Is a Steerable Concept Representation Real? Measurement Confounds in a Cross-Family Audit of Neuroscience Parallels in LLMs

ArXiv CS.AI **50%**

arXiv:2608.08159v1 Announce Type: new Abstract: Large language models (LLMs) are increasingly reported to exhibit human-like neural and cognitive signatures, including concept cells, mental number lines, and cognitive maps. These claims often rely on linear probing and...

https://arxiv.org/abs/2608.08159

5 CONF

Agentic AI-driven Immersive Simulation: A Knowledge-Aware Virtual Training Platform forHigh Dose Rate (HDR) Brachytherapy

ArXiv CS.AI 50%

arXiv:2608.08163v1 Announce Type: new Abstract: The convergence of the Metaverse and Large Language Model (LLM)-based AI agent is catalyzing a shift toward autonomous, immersive, and personalized pedagogical frameworks in medical education. This paper presents a novel agentic...

https://arxiv.org/abs/2608.08163

6 CONF

Large Multimodal Agents for Intelligent Transportation Systems: Architectures, Evidence, and Deployment Challenges

ArXiv CS.AI 50%

arXiv:2608.08184v1 Announce Type: new Abstract: Large multimodal agents (LMAs) are increasingly proposed for intelligent transportation systems (ITS), but existing studies often conflate multimodality, agency, empirical performance, and deployment readiness. This review provides...

https://arxiv.org/abs/2608.08184

7 CONF

Janus: An Algorithm-Evaluator Co-Evolution Framework for LLM-Driven Discovery under Expensive Evaluation Budgets

ArXiv CS.AI 50%

arXiv:2608.08189v1 Announce Type: new Abstract: LLM-driven program discovery relies on rapid evaluator feedback, but many scientific and engineering tasks require high-fidelity simulations, hardware execution, or physical experiments, making each evaluation expensive. Cheap...

https://arxiv.org/abs/2608.08189

8 CONF

Automated Generation of Complexity-Validated Decision Scenarios Using Large Language Models

ArXiv CS.AI 50%

arXiv:2608.08822v1 Announce Type: new Abstract: Cognitive decision-making research depends on diverse scenarios with carefully controlled complexity, yet manual production is slow, inconsistent, and biased. We developed an automated pipeline that uses LLms to generate structured...

https://arxiv.org/abs/2608.08822

9 CONF

Illusion of Alignment: Detecting Hidden Disagreement in Collaborative Dialogue

ArXiv CS.AI 50%

arXiv:2608.08210v1 Announce Type: new Abstract: Collaborative dialogue can end with apparent agreement while participants still differ on goals, assumptions, or execution plans, creating an \textbf{illusion of alignment (IoA)}. A real-user study across 18 meetings confirms that...

https://arxiv.org/abs/2608.08210

0 CONF

Harmful Content Is Not Enough: Continuation Framing Moderates In-Context Emergent Misalignment

ArXiv CS.AI 50%

arXiv:2608.08212v1 Announce Type: new Abstract: In-context learning (ICL) can induce emergent misalignment (EM), where narrow misaligned examples alter answers to unrelated questions. Existing prompts, however, conflate harmful-text exposure with an invitation to continue...

https://arxiv.org/abs/2608.08212

1 CONF

A Fair Objective for Human-Empowerment-Preserving AI: Desiderata, Design, and Likely Behavioral Consequences

ArXiv CS.AI 50%

arXiv:2608.08240v1 Announce Type: new Abstract: This paper explores the idea of promoting well-being and safety in human-AI interactions by forcing AI agents explicitly to empower humans and to manage the power balance between humans and AI agents in a desirable way. Using a...

https://arxiv.org/abs/2608.08240

2 CONF

SuperLocalMemory 4.0: The Governed Memory Operating System for AI Agents

ArXiv CS.AI 50%

arXiv:2608.08253v1 Announce Type: new Abstract: AI agents are becoming shared infrastructure, yet durable memory is commonly assembled from separate retrieval, governance, and operational components. We present SuperLocalMemory 4.0, a governed, local-first memory operating...

https://arxiv.org/abs/2608.08253

3 CONF

Your Prompt Is Not the Only Prompt: How Much Do LLMs Weight Structured-Output Schema Descriptions?

ArXiv CS.AI 50%

arXiv:2608.08254v1 Announce Type: new Abstract: Structured output, where an LLM populates a predefined JSON schema, has become a default mechanism for data labeling and information extraction, but it also introduces a second instruction channel through schema descriptions. We...

https://arxiv.org/abs/2608.08254

4 CONF

Exploring LLM Capabilities for Situational Understanding and COLREG compliance on real-world maritime navigation scenarios

ArXiv CS.AI 50%

arXiv:2608.08281v1 Announce Type: new Abstract: Recently, Large Language Models (LLMs) have shown considerable capability for situational understanding, reasoning, and decision making in different domains, most notable in the automotive sector. Therefore, we explore current...

https://arxiv.org/abs/2608.08281

5 CONF

Quokka: Accelerating Program Verification with LLMs via Invariant Synthesis

ArXiv CS.AI 50%

arXiv:2509.21629v4 Announce Type: replace-cross Abstract: Program verification relies on loop invariants, yet automatically discovering strong invariants remains a long-standing challenge. We investigate whether large language models (LLMs) can accelerate program verification by...

https://arxiv.org/abs/2509.21629

6 CONF

SuperNeuroMAT: An Efficient Matrix-based Simulator for Spiking Neural Networks

ArXiv CS.AI 50%

arXiv:2608.08479v1 Announce Type: cross Abstract: Spiking neural networks (SNNs) offer a promising pathway to energy-efficient AI and brain-inspired computing. However, their widespread adoption is hindered by a lack of fast, accessible, and versatile simulation frameworks. In...

https://arxiv.org/abs/2608.08479

7 **CONF**

DoGMA: A Central-Dogma-Guided Foundation Model for Multi-Omics Alignment and Multi-Task Learning in Oncology

ArXiv CS.AI **50%**

arXiv:2608.08148v1 Announce Type: cross Abstract: Attention mechanisms have been widely utilized in modern deep learning, and many existing multi-omics models inherit their conventional use to allow unrestricted bidirectional interactions. However, the fundamental logic of life...

https://arxiv.org/abs/2608.08148

8 **CONF**

CAP: A Scalable Benchmark for Evaluating Cross-Site Browser Agents with Complex Actions and Perception

ArXiv CS.AI **50%**

arXiv:2608.08392v1 Announce Type: new Abstract: Large language models are increasingly deployed as autonomous agents that interact with the web through browsers. While recent progress has been driven by benchmarks that evaluate end-to-end task success, these evaluations largely...

https://arxiv.org/abs/2608.08392

9 **CONF**

Forgotten History or Test-of-Time? Retrospect and Prospect on RAG from an IR Perspective

ArXiv CS.AI **50%**

arXiv:2608.08445v1 Announce Type: new Abstract: Retrieval-Augmented Generation (RAG) is widely regarded as a novel paradigm born from the limitations of large language models (LLMs)--a mechanism to ground their outputs in external knowledge. This view, however, is incomplete...

https://arxiv.org/abs/2608.08445

0 **CONF**

LLM within MCP Matters: Measuring Inefficient Resource Utilization Driven by LLMs

ArXiv CS.AI **50%**

arXiv:2608.08467v1 Announce Type: new Abstract: The Model Context Protocol (MCP) standardizes how servers expose data and tools to Large Language Models (LLMs). A common server design embeds frequently used reference data, such as identifier lookup tables, directly in the server...

https://arxiv.org/abs/2608.08467

1 **CONF**

Yesterday's Shield, Today's Spear: A Self-Evolving Safety Guardrail in Production

ArXiv CS.AI **50%**

arXiv:2608.08471v1 Announce Type: new Abstract: Deployed LLM safety guardrails are predominantly static: trained once and frozen at release, while new jailbreak techniques and previously un-addressed harmful categories emerge within days, leaving the defense perpetually a step...

https://arxiv.org/abs/2608.08471

2 **CONF**

HoloAegis: Frozen Representation, Topological Inference: Minimally Parametric Safety Manifolds for Zero-Shot LLM Guardrails

ArXiv CS.AI **50%**

arXiv:2608.08485v1 Announce Type: new Abstract: Current LLM safety guardrails face a fundamental tension: fine-tuning distorts pre-trained representations while generative judges incur prohibitive inference costs. We challenge the prevailing paradigm by asking: can safety be...

https://arxiv.org/abs/2608.08485

3 **CONF**

SymboUQ: Symbolic Uncertainty Quantification for Spatial Reasoning in LLMs

ArXiv CS.AI 50%

arXiv:2608.00417v2 Announce Type: replace Abstract: Although large language models (LLMs) can produce fluent spatial reasoning traces, their intermediate relations may fail to support the final conclusion, making token-level confidence insufficient for final-answer reliability...

https://arxiv.org/abs/2608.00417

4 **CONF**

Understanding Reasoning from Pretraining to Post-Training

ArXiv CS.AI 50%

arXiv:2607.16097v2 Announce Type: replace-cross Abstract: Reinforcement learning (RL) has become central to improving large language models (LLMs) on complex reasoning tasks, yet RL post-training is largely studied in isolation from the pretraining that precedes it. As a result,...

https://arxiv.org/abs/2607.16097

5 **CONF**

Discovering Diverse Planning Policies for Multimodal Embodied Agents with Quality-Diversity Optimization

ArXiv CS.AI 50%

arXiv:2608.08523v1 Announce Type: new Abstract: Multimodal embodied agents are increasingly required to solve long-horizon tasks by integrating visual observations, textual goals, and interaction history into closed-loop decision making. However, state-of-the-art...

https://arxiv.org/abs/2608.08523

6 **CONF**

VoxZip: Semantic-Anchored Temporal KV Cache Compression for Long-Context Audio Inference

ArXiv CS.AI 50%

arXiv:2608.08569v1 Announce Type: new Abstract: Recent advancements in Speech Large Language Models have demonstrated remarkable capabilities in understanding complex audio tasks. Despite this progress, their long-context inference remains severely bottlenecked by prohibitive KV...

https://arxiv.org/abs/2608.08569

7 **CONF**

FailForge: Distilling Procedural Competence from Persistent Failures into Code Agents

ArXiv CS.AI 50%

arXiv:2608.08570v1 Announce Type: new Abstract: Rejection sampling fine-tuning (RFT) is widely used to train code agents by generating trajectories on verifiable software engineering tasks, retaining those that pass the tests, and fine-tuning on the successful rollouts. However,...

https://arxiv.org/abs/2608.08570

8 **CONF**

Unaccountable Delegation, Fading Skills: Mapping the Risks of Workplace AI Agents

ArXiv CS.AI 50%

arXiv:2608.08601v1 Announce Type: new Abstract: To anticipate socio-technical risks from AI agents, organizations need taxonomies to classify them. However, existing AI risk taxonomies focus on broad risks and do not capture job-specific risks introduced by agents. To address...

https://arxiv.org/abs/2608.08601

9 **CONF**

MMArch: Benchmarking Multimodal Reasoning Grounded in Architectural Evidence

ArXiv CS.AI 50%

arXiv:2608.09281v1 Announce Type: new Abstract: Multimodal large language models (MLLMs) perform strongly on engineering imagery, yet existing benchmarks mostly test drawing recognition, information extraction, or compliance checking, leaving open whether models can combine...

https://arxiv.org/abs/2608.09281

0 **CONF**

A QUBO-Inspired Computational Framework for Airport Landside Bottleneck Diagnosis and Dynamic Dispatch Optimization

ArXiv CS.AI 50%

arXiv:2608.08632v1 Announce Type: new Abstract: Airport landside traffic centers connect terminal arrivals with taxis, ride-hailing vehicles, private cars, buses, metro services, parking facilities, and terminal-area roadways. Peak arrivals can create coupled congestion across...

https://arxiv.org/abs/2608.08632

1 **CONF**

The Scaffolding Matters More Than the Interface: A Controlled Comparison of MCP and CLI Tool Use Across Seven Agent Scaffoldings, Five Language Models, and One Software Task

ArXiv CS.AI 50%

arXiv:2608.08654v1 Announce Type: new Abstract: How much an AI coding agent costs to run can depend more on the agent scaffolding that drives it than on the interface through which it reaches its tools. We set out to measure the cost of tool use over the Model Context Protocol...

https://arxiv.org/abs/2608.08654

2 **CONF**

A Structural Dynamics Graph World Model: Unified Modeling, Constrained Rollout, and Interpretable Calibration

ArXiv CS.AI 50%

arXiv:2608.08689v1 Announce Type: new Abstract: The state evolution of a complex system arises jointly from object laws, relational propagation, domain conservation, and unmodeled error. Forcing all sources into one black box makes mechanism attribution and constraint...

https://arxiv.org/abs/2608.08689

3 **CONF**

EnergyBridge: Benchmarking Household Energy Management, User Participation, and Grid Flexibility

ArXiv CS.AI 50%

arXiv:2608.08691v1 Announce Type: new Abstract: Residential virtual power plants (VPPs) can provide grid flexibility by shifting household demand, but physical flexibility becomes dependable capacity only when residents authorize a plan and the promised response is delivered....

https://arxiv.org/abs/2608.08691

4 **CONF**

AI Evaluation Should Measure Verification Cost, Not Correctness Alone

ArXiv CS.AI 50%

arXiv:2608.08709v1 Announce Type: new Abstract: The reliability of AI generative models is typically measured by output correctness, yet in practice it depends on the effort required to verify those outputs. We argue that current evaluation metrics overlook a critical failure...

https://arxiv.org/abs/2608.08709

5 CONF

Scale-to-Dialogue: Low-Burden Elicitation of Daily Premenstrual Symptom Ratings with Small Language Models

ArXiv CS.AI 50%

arXiv:2608.08746v1 Announce Type: new Abstract: Prospective daily symptom tracking is central to premenstrual health assessment, but repeated ordinal forms impose substantial response burden. We formulate conversational administration as an ordinal label-recovery problem: the...

https://arxiv.org/abs/2608.08746

6 CONF

SymDiag: Explainable Diagnosis for LLM Reasoning via Neuro-Symbolic Verification

ArXiv CS.AI 50%

arXiv:2608.08786v1 Announce Type: new Abstract: Large language models (LLMs) increasingly serve as data-driven reasoners, yet their chains-of-thought (CoT) can be unfaithful even when final answers are correct. Most existing ``verification'' signals are not diagnostic: answer...

https://arxiv.org/abs/2608.08786

7 CONF

Listen, See and Track: Spatio-Temporal Audio-Visual Sound Event Reasoning for Omni-Modal Language Models

ArXiv CS.AI 50%

arXiv:2608.09435v1 Announce Type: new Abstract: Understanding dynamic sound sources requires jointly determining what produces a sound, where the source is located, and how it moves over time. Yet existing audio-language models often represent clips as global acoustic events,...

https://arxiv.org/abs/2608.09435

8 CONF

Findings of the First Teaching Monster Challenge: A Benchmark of Pedagogical Content Knowledge in AI Agents

ArXiv CS.AI 50%

arXiv:2608.08852v1 Announce Type: new Abstract: AI agents can now solve problems, answer like subject experts, and generate long-form multimodal content. However, whether they can adapt a lesson to fit a specified learner, which education calls Pedagogical Content Knowledge...

https://arxiv.org/abs/2608.08852

9 CONF

Application of Artificial Intelligence for Fraudulent Banking Operations Recognition

ArXiv CS.AI 50%

arXiv:2608.07471v1 Announce Type: cross Abstract: This study considers the task of applying artificial intelligence to recognize bank fraud. In recent years, due to the COVID19 pandemic, bank fraud has become even more common due to the massive transition of many operations to...

https://arxiv.org/abs/2608.07471

00 CONF

AquiLLM: An Architecture for Supporting Tacit Knowledge Capture in Research Groups

ArXiv CS.AI 50%

arXiv:2608.08883v1 Announce Type: new Abstract: Recent advances in retrieval-augmented generation (RAG) and large language models (LLMs) enable researchers to integrate AI into scientific workflows. However, using proprietary commercial AI systems raises concerns about...

https://arxiv.org/abs/2608.08883

01 CONF

Full-bandwidth transformer

ArXiv CS.AI 50%

arXiv:2608.08888v1 Announce Type: new Abstract: Autoregressive transformers compute along two axes: horizontally across generated tokens, and vertically through model depth. Dense attention gives each token broad horizontal access to the past, but the vertical feedback channel...

https://arxiv.org/abs/2608.08888

02 CONF

LLM Reasoning for Subjective Tasks: Failure Modes, Mitigation, and Dynamic Reasoning Routing

ArXiv CS.AI 50%

arXiv:2608.08889v1 Announce Type: new Abstract: Recommendation systems thrive on personalization, where ''correctness'' is rarely a binary truth but a matter of subjective human preference. As Large Language Models (LLMs) are deployed as autonomous verifiers of safety and...

https://arxiv.org/abs/2608.08889

03 CONF

From Manuals to Maintenance: Fine-Tuning MedGemma for Multi-Modal Imaging System Support in Low-Resource Settings

ArXiv CS.AI 50%

arXiv:2608.08896v1 Announce Type: new Abstract: Imaging device downtime is a major barrier to healthcare delivery in low- and middle-income countries (LMICs), often driven by limited access to specialized biomedical engineering support. We present a multi-modality medical...

https://arxiv.org/abs/2608.08896

04 CONF

Expert-Guided Multimodal Fusion for Unified Emotion and Sentiment Analysis

ArXiv CS.AI 50%

arXiv:2601.07565v2 Announce Type: replace-cross Abstract: Multimodal emotion understanding requires the integration of heterogeneous data sources, including text, audio, and visual modalities, while simultaneously addressing discrete emotion recognition and continuous sentiment...

https://arxiv.org/abs/2601.07565

05 CONF

Not an A11y: How Android Accessibility Exposes Mobile AI Agents to Indirect Prompt Injection

ArXiv CS.AI 50%

arXiv:2608.08939v1 Announce Type: new Abstract: The rise of autonomous AI agents represents a major paradigm shift in how users interact with mobile devices. Frameworks such as MobileRun and Mobile-Use can autonomously navigate Android applications and execute complex multi-step...

https://arxiv.org/abs/2608.08939

06 CONF

Multi-Agent AI Safety as an Institutional Design Problem

ArXiv CS.AI 50%

arXiv:2608.09828v1 Announce Type: cross Abstract: AI agents increasingly work inside systems that govern how they delegate tasks, move information, execute actions, and use shared resources. Recent work already shows that deployment rules can change collective behavior. Here we...

https://arxiv.org/abs/2608.09828

07 CONF

DualCert: A Solver for the Traveling Salesman Problem with Constraint-Coupled Learning

ArXiv CS.AI **50%**

arXiv:2608.09042v1 Announce Type: new Abstract: Large traveling salesman problem (TSP) instances require a solver to allocate limited computation while preserving the validity of its outputs. Existing neural--operations-research (OR) hybrids predict guidance without requiring...

https://arxiv.org/abs/2608.09042

08 CONF

Different Feedback, Different Updates: Selective Self-Learning from User Interactions for Large Language Models

ArXiv CS.AI **50%**

arXiv:2608.09109v1 Announce Type: new Abstract: User feedback offers natural supervision for persistent LLM improvement, but a single message may support multiple behavioral changes with different scopes of generalization. We introduce SLIFT, a selective self-learning framework...

https://arxiv.org/abs/2608.09109

09 CONF

ChronoState: Hidden Elapsed-Time Conditioning for Temporal-State Action Selection in Frozen-Backbone Language Models

ArXiv CS.AI **50%**

arXiv:2608.09124v1 Announce Type: new Abstract: Temporal decisions in language-model systems often depend on both symbolic task state and elapsed wall-clock time, such as cache expiration, job completion, quota resets, deadlines, or stale sessions. We study whether elapsed time...

https://arxiv.org/abs/2608.09124

10 CONF

CIDER: A Dataset of Contextual Disclosure Boundaries for Privacy Preference Alignment

ArXiv CS.AI **50%**

arXiv:2608.09164v1 Announce Type: new Abstract: Aligning large language models (LLMs) with human privacy preferences requires capturing individuals' disclosure boundaries beyond general privacy norms. However, a gap remains in eliciting such nuanced preferences to evaluate...

https://arxiv.org/abs/2608.09164

11 CONF

From Relevance to Execution Utility: Reward-Aware Dynamic Execution Gating for Skill-Based LLM Agents

ArXiv CS.AI **50%**

arXiv:2608.09168v1 Announce Type: new Abstract: Agent skills are increasingly used to equip large language model (LLM) agents with reusable procedural knowledge. Although recent work has substantially improved skill retrieval due to the increasing skill libraries, retrieving a...

https://arxiv.org/abs/2608.09168

12 CONF

Structure-Preserving Uncertainty Propagation in First-Order Proof Search

ArXiv CS.AI **50%**

arXiv:2608.09190v1 Announce Type: new Abstract: GK is a query-directed first-order prover that extends ordinary resolution-based proof search with explicit positive and negative claims, numerical confidence values, and prioritized default rules with exceptions. It works directly...

https://arxiv.org/abs/2608.09190

13 **CONF**

CRUISE: Vision-Language Model-Guided Uncertainty-Aware Cross-Modal Sensor Fusion for Robust Autonomous Driving

ArXiv CS.AI 50%

arXiv:2608.09202v1 Announce Type: new Abstract: Modern autonomous vehicles are equipped with multiple sensors, such as cameras, LiDAR, and radar, for comprehensive environmental perception. However, robust cross-modal feature fusion remains a critical challenge, as the...

https://arxiv.org/abs/2608.09202

14 **CONF**

Omni2LoRA: Coherence-Preserving Parametric Memory for Efficient Omni Language Models

ArXiv CS.AI 50%

arXiv:2608.09227v1 Announce Type: new Abstract: Omnimodal language models (OLMs) enable unified audio-visual understanding, but processing long joint token sequences makes inference computationally prohibitive. While recent token compression methods attempt to alleviate this...

https://arxiv.org/abs/2608.09227

15 **CONF**

MCIF: Multimodal Crosslingual Instruction-Following Benchmark from Scientific Talks

ArXiv CS.AI 50%

arXiv:2507.19634v4 Announce Type: replace-cross Abstract: Recent advances in large language models have laid the foundation for multimodal LLMs (MLLMs), which unify text, speech, and vision within a single framework. As these models are rapidly evolving toward general-purpose...

https://arxiv.org/abs/2507.19634

16 **CONF**

Tools to Explain Neural Networks for Power System Dynamics

ArXiv CS.AI 50%

arXiv:2608.08048v1 Announce Type: cross Abstract: This paper presents, for the first time in power systems literature to our knowledge, analytical tools to explain the training performance of machine learning surrogate models for power system dynamics. Power system simulations...

https://arxiv.org/abs/2608.08048

17 **CONF**

ASPaeroFlow: Decomposition Heuristics for Joint Air Traffic Flow & Capacity Management

ArXiv CS.AI 50%

arXiv:2608.09315v1 Announce Type: new Abstract: While mathematical models act as vital decision support systems for operational Air Traffic Flow and Capacity Management (ATFCM), existing approaches isolate Air Traffic Flow Management (ATFM) from Dynamic Airspace Configuration...

https://arxiv.org/abs/2608.09315

18 **CONF**

Ground-Truth Neighborhood Regularization for Reinforcement Learning Post-Training of Time Series Foundation Models

ArXiv CS.AI 50%

arXiv:2608.08010v1 Announce Type: cross Abstract: Time series forecasting (TSF) plays an important role in a wide range of real-world applications. Recently, time series foundation models (TSFMs), pretrained on large-scale datasets, have demonstrated strong generalization...

https://arxiv.org/abs/2608.08010

19 **CONF**

LLM-Guided Heuristic Design from Simulation Traces: A Case Study in Dynamic Production and AGV Scheduling

ArXiv CS.AI **50%**

arXiv:2608.09343v1 Announce Type: new Abstract: Simulation-based optimization (SBO) evaluates executable policies under stochastic dynamics, but most methods treat the simulator as a black box: aggregate scores rank candidates without revealing why they fail or which policy...

https://arxiv.org/abs/2608.09343

20 **CONF**

Capability Is Not Propensity: Measuring Pressure-Robust Cooperative Behavior in Civic LLM Agents

ArXiv CS.AI **50%**

arXiv:2608.09485v1 Announce Type: new Abstract: Cooperative capabilities in language models are dual-use. The same social reasoning that supports civic deliberation can also enable strategic omission, false consensus, and manipulative framing. We argue that Cooperative AI...

https://arxiv.org/abs/2608.09485

21 **CONF**

The Politician, the Liar, and the Obedient Worker: Emerging Behavior of LLM Agents in Hierarchical Games

ArXiv CS.AI **50%**

arXiv:2608.09574v1 Announce Type: new Abstract: LLMs are rapidly embedding themselves into daily life: drafting our emails, managing our schedules, and making decisions on our behalf. As they move from individual tools to participants in multi-agent organizations, an important...

https://arxiv.org/abs/2608.09574

22 **CONF**

ElasticBack: Stealthy Conditional Backdoor in LLM-Agent Skills via Coupled Trigger-Rule Optimization

ArXiv CS.AI **50%**

arXiv:2608.09577v1 Announce Type: new Abstract: Agent skills, bundles of instructions and resources that an LLM agent loads on demand, form an emerging supply chain where a single poisoned skill can persistently compromise every agent that installs it. However, existing skill...

https://arxiv.org/abs/2608.09577

23 **CONF**

CoRCi: Cross-Reconstruction of Coherent Interests Modeling in Cross-Domain Sequential Recommendation

ArXiv CS.AI **50%**

arXiv:2608.09580v1 Announce Type: new Abstract: Cross-Domain Sequential Recommendation (CDSR) aims to alleviate data sparsity by transferring dynamic user interests across related domains. A key challenge lies in effectively bridging these domains. In single-domain modeling,...

https://arxiv.org/abs/2608.09580

24 **CONF**

ICM Out! Better Tournament Strategy from Computed Continuations, vs. Solvers and LLMs

ArXiv CS.AI **50%**

arXiv:2608.09586v1 Announce Type: new Abstract: The Independent Chip Model (ICM) converts tournament chips into reference prize equity, and policies are routinely constructed against those values. Because ICM reads only stack sizes, it omits action order, blind obligations, and...

https://arxiv.org/abs/2608.09586

25 CONF

From Sweep to Seam: Interleaved Cross-Block Post-Training Quantization

ArXiv CS.AI 50%

arXiv:2608.09595v1 Announce Type: new Abstract: Compressing large language models to two bits or fewer is increasingly feasible through block-wise post-training quantization; cross-block variants reconstruct neighboring Transformer blocks within a moving window. In the fixed...

https://arxiv.org/abs/2608.09595

26 CONF

Avalon-ToM-Bench: Evaluating Fine-Grained Theory of Mind via Asymmetric Game Mechanics

ArXiv CS.AI 50%

arXiv:2608.09638v1 Announce Type: new Abstract: Theory of Mind (ToM) is essential for agent interactions, yet existing evaluations either rely on static scenarios that oversimplify mental-state reasoning or interactive settings that provide limited diagnostic insight. We present...

https://arxiv.org/abs/2608.09638

27 CONF

Model Discovery Agent: LLM-assisted Bayesian experiment design for data-efficient discovery of mechanistic world models

ArXiv CS.AI 50%

arXiv:2608.09696v1 Announce Type: new Abstract: Predicting the answer to interventional ``what if'' questions --- the outcome of an action never taken --- requires a \emph{mechanistic}, causal model, not a curve fit; and learning such a model requires \emph{experiments}, because...

https://arxiv.org/abs/2608.09696

28 CONF

AirFlow: Context Preserving and Multi-Rate State Modeling for Air Quality Forecasting

ArXiv CS.AI 50%

arXiv:2608.09775v1 Announce Type: new Abstract: Accurate air quality forecasting is essential for public health and urban environmental management, but remains challenging because pollutant channels differ in periodicity and distribution drift, while their concentration...

https://arxiv.org/abs/2608.09775

29 CONF

SkillConsist: Detecting Inconsistencies in Agent Skills via Bidirectional Graph Alignment

ArXiv CS.AI 50%

arXiv:2608.07639v1 Announce Type: cross Abstract: Agent Skills provide reusable capabilities to LLM agents. Agent Skill inconsistencies can expose undisclosed dangerous behavior or cause wrong Skill selection. Recent Agent Skill research has increasingly examined Agent Skill...

https://arxiv.org/abs/2608.07639

30 CONF

Positioning Generative Artificial Intelligence in STEM Assessment: When to Require, Scaffold, or Restrict Its Use

ArXiv CS.AI 50%

arXiv:2608.07475v1 Announce Type: cross Abstract: Generative Artificial Intelligence (GenAI) presents a governance challenge for STEM assessment. Unrestricted access can enable task outsourcing that undermines the validity of traditional assessments, while blanket prohibitions...

https://arxiv.org/abs/2608.07475

31 **CONF**

Designing for Ethical AI: HCI Feature Considerations to Improve Fairness and User Experience in AutoML use for Human Resources

ArXiv CS.AI 50%

arXiv:2608.07477v1 Announce Type: cross Abstract: This thesis examines the fairness of Automated Machine Learning (AutoML) tools in human resource hiring systems through the combined lenses of regulation, business strategy, and Human-Computer Interaction (HCI). It argues that...

https://arxiv.org/abs/2608.07477

32 **CONF**

EmoPatient: An Emotion-Directed Patient Simulator for Realistic Palliative Care Communication Training

ArXiv CS.AI 50%

arXiv:2608.07495v1 Announce Type: cross Abstract: Effective communication during palliative care discussions is a critical clinical skill, yet training clinicians to manage complex patient emotions remains challenging. Large language model (LLM)-based patient simulators provide...

https://arxiv.org/abs/2608.07495

33 **CONF**

Knowing You Is Everything: LLM Agents Achieve Near-Perfect Profile-Consistent Reaction Prediction in Social Media Simulation

ArXiv CS.AI 50%

arXiv:2608.07498v1 Announce Type: cross Abstract: Autonomous AI agents in social media present concrete risks to democratic discourse and platform governance, while also offering tools for pre-deployment recommender system testing. A central open question is whether...

https://arxiv.org/abs/2608.07498

34 **CONF**

Evaluation of Motivational Interviewing Counsellors with Task-Aware Multi-Stage LLM-Based Simulated Clients

ArXiv CS.AI 50%

arXiv:2608.07499v1 Announce Type: cross Abstract: The development and benchmarking of Large Language Model (LLM)-based Motivational Interviewing (MI) counsellors now often rely on LLM-based simulated clients. Prior work on simulated clients, however, has not aligned with the...

https://arxiv.org/abs/2608.07499

35 **CONF**

The Scaling Paradox in Human-AI Collaboration

ArXiv CS.AI 50%

arXiv:2608.00818v2 Announce Type: replace Abstract: The discovery of scaling laws has highlighted the extraordinary potential of AI systems with a striking empirical pattern: as AI systems scale, their capabilities tend to improve predictably. Yet, in real-world applications, AI...

https://arxiv.org/abs/2608.00818

36 **CONF**

JaleesBench: Are AI Assistants Good Spiritual Company?

ArXiv CS.AI 50%

arXiv:2608.07508v1 Announce Type: cross Abstract: Large language models are already advisors to millions of people of faith who bring them real decisions. The pressing question for a person of faith is not what a model knows or professes but what its counsel does to the person...

https://arxiv.org/abs/2608.07508

37 CONF

Frequency-Domain Dual-Branch Fusion for Medical Visual Question Answering

ArXiv CS.AI 50%

arXiv:2608.08307v1 Announce Type: cross Abstract: Medical Visual Question Answering (VQA) requires aligning subtle visual evidence, including lesion texture, boundary sharpness, and diffuse density changes, with clinical language. Existing multimodal fusion approaches operating...

https://arxiv.org/abs/2608.08307

38 CONF

KumbhDoot: A Scale-Ready, LLM-Bounded Architecture for Mass-Gathering Public-Service Assistants

ArXiv CS.AI 50%

arXiv:2608.07520v1 Announce Type: cross Abstract: Mass religious gatherings such as the Kumbh Mela concentrate tens of millions of people into a single region over a few weeks, producing intense, repetitive, multilingual, and safety-critical demand for information. The default...

https://arxiv.org/abs/2608.07520

39 CONF

From Evaluated Models to Evaluation Aids: A Multi-Evidence Study of LLM-Based Difficulty Calibration for Programming Examinations

ArXiv CS.AI 50%

arXiv:2608.07523v1 Announce Type: cross Abstract: Difficulty differences across parallel-class programming examinations affect the fairness of course assessment. This study repositions large language models from benchmark evaluation targets to auxiliary evidence sources for...

https://arxiv.org/abs/2608.07523

40 CONF

Evolving Safety Landscape of Multi-modal Large Language Models: A Survey of Emerging Threats and Safeguards

ArXiv CS.AI 50%

arXiv:2608.07535v1 Announce Type: cross Abstract: Multi-modal large language models (MLLMs) integrate heterogeneous modalities through modality alignment and fusion, enabling stronger understanding and reasoning. However, this architectural shift reshapes the safety landscape of...

https://arxiv.org/abs/2608.07535

41 CONF

MOSAIC: Adversarial Co-evolution of Specialist Heuristics and Problem Instances for LLM-based Automated Heuristic Design

ArXiv CS.AI 50%

arXiv:2608.07544v1 Announce Type: cross Abstract: Automated heuristic design (AHD) with large language models (LLMs) has produced strong heuristics for combinatorial optimization problems (COPs). Yet existing frameworks optimize for average performance on a small fixed dataset...

https://arxiv.org/abs/2608.07544

42 CONF

Generalizing deep reinforcement learning across cable-driven parallel robot configurations with actuator-level policies

ArXiv CS.AI 50%

arXiv:2608.07546v1 Announce Type: cross Abstract: Cable-driven parallel robots (CDPRs) present diverse configurations and complex control challenges, which can be addressed by deep reinforcement learning (DRL) by learning their nonlinear dynamics. However, DRL methods often...

https://arxiv.org/abs/2608.07546

43 **CONF**

Learning an Interior Layout Policy in a Domain Specific Language Action Space

ArXiv CS.AI 50%

arXiv:2608.07547v1 Announce Type: cross Abstract: Indoor scene layout generation is a challenging task in interior design. Existing methods often oversimplify the task by reducing room conditions to coarse 3D bounding boxes and neglecting structural elements such as doors and...

https://arxiv.org/abs/2608.07547

44 **CONF**

MasDrift: Benchmarking Authorization Preservation Across Multi-Agent Architectures

ArXiv CS.AI 50%

arXiv:2608.07556v1 Announce Type: cross Abstract: Multi-agent systems (MAS) decompose long-horizon tasks across supervisors and subagents, but delegated goals do not necessarily carry their original authorization boundaries. Existing safety benchmarks mainly study adversarial...

https://arxiv.org/abs/2608.07556

45 **CONF**

Temporal Generalization in fNIRS-Based Autism Classification: A Cross-Time-Window Transfer Benchmark

ArXiv CS.AI 50%

arXiv:2608.07567v1 Announce Type: cross Abstract: Functional near-infrared spectroscopy (fNIRS) is a promising modality for autism spectrum disorder (ASD) classification, yet existing approaches assume temporally aligned evaluation. In practice, the optimal observation window...

https://arxiv.org/abs/2608.07567

46 **CONF**

BRACE: Taming Sharp Irregularities via Barycentric Rational Forecasting for Fast Diffusion Transformers Inference

ArXiv CS.AI 50%

arXiv:2608.07572v1 Announce Type: cross Abstract: Diffusion Transformers (DiTs) have demonstrated exceptional performance in high-fidelity image and video generation. To alleviate their massive computational overhead, temporal feature caching has been proposed to bypass...

https://arxiv.org/abs/2608.07572

47 **CONF**

Weather- and Location-Aware Agentic Dining Recommendation: Leveraging LLM World Knowledge for Region-Sensitive Contextual Reasoning

ArXiv CS.AI 50%

arXiv:2608.07593v1 Announce Type: cross Abstract: Context-aware recommender systems have long recognized that factors such as location, time, and weather shape where and what people choose to eat. Existing weather-aware food and point-of-interest recommenders, however, typically...

https://arxiv.org/abs/2608.07593

48 **CONF**

Hit Selection Using SSMD-Based Machine Learning Performance Metrics in High-Throughput Screening Assays

ArXiv CS.AI 50%

arXiv:2608.07609v1 Announce Type: cross Abstract: High-throughput screening (HTS) assays are central to early-stage drug discovery but are often limited by extreme data sparsity, as primary screens typically use only a single replicate per test substance. This sparsity makes...

https://arxiv.org/abs/2608.07609

49 **CONF**

Complete, Scalable, and Robust Prioritized Planning for Multi-Robot Ordered Storage and Retrieval at Maximum Capacity

ArXiv CS.AI 50%

arXiv:2608.07734v1 Announce Type: cross Abstract: Automated warehouses face a fundamental trade-off between maximizing storage density and achieving high retrieval throughput. While puzzle-based storage (PBS) architectures increase capacity by eliminating aisles, coordinating...

https://arxiv.org/abs/2608.07734

50 **CONF**

CFD-Guided Detection of Concept Drift in Multimodal Physiologic Signals

ArXiv CS.AI 50%

arXiv:2608.07759v1 Announce Type: cross Abstract: Cardiovascular AI models can classify clean elec- trocardiogram (ECG) signals, but real wearable signals change because of motion, breathing, posture, sensor contact, and true clinical deterioration. This paper asks when a model...

https://arxiv.org/abs/2608.07759

51 **CONF**

Illusion or Integrity? Geometrical Consistency Metric for AIGC Video Quality Evaluation

ArXiv CS.AI 50%

arXiv:2608.09594v1 Announce Type: cross Abstract: Recently, AI-driven video generation has attracted considerable attention. This surge increases the demand for reliable video quality assessment (VQA) metrics to evaluate AI-generated content (AIGC) videos and guide model...

https://arxiv.org/abs/2608.09594

52 **CONF**

Router Sensitivity Under Lightweight Fine-Tuning Identifies Prunable Experts in Mixture-of-Experts Models

ArXiv CS.AI 50%

arXiv:2608.07890v1 Announce Type: cross Abstract: Mixture-of-Experts (MoE) models decouple total parameters from per-token compute, but deployment still requires storing every expert. Recent theory shows that pruning experts with the smallest router-norm changes during...

https://arxiv.org/abs/2608.07890

53 **CONF**

Beyond "I Can't Help With That": How Child Safety Experts Evaluate AI Chatbot Safety

ArXiv CS.AI 50%

arXiv:2608.07902v1 Announce Type: cross Abstract: Youth increasingly turn to AI chatbots for social and emotional support, raising concerns about how these systems respond, especially in high-stakes situations. However, existing child safety evaluations of AI lack grounding in...

https://arxiv.org/abs/2608.07902

54 **CONF**

FeedbackTrack: Visual-Cortex-Inspired Cross-Frame Feedback for Transformer Tracking

ArXiv CS.AI 50%

arXiv:2608.09369v1 Announce Type: cross Abstract: Visual object tracking requires effective temporal integration, yet most Transformer trackers still rely on predominantly feed-forward feature extraction. Existing temporal mechanisms typically update templates, prompts, queries,...

https://arxiv.org/abs/2608.09369

55 **CONF**

Persistent Semantic Entities in Tool-Augmented LLM Systems

ArXiv CS.AI **50%**

arXiv:2608.07952v1 Announce Type: cross Abstract: Tool-augmented LLM agents can harbor implicit state that persists across sessions, activates through events, and propagates across agent boundaries---largely invisible to standard debugging. We formalize this as Persistent...

https://arxiv.org/abs/2608.07952

56 **CONF**

EasyBalance: Cross-Layer Load Balancing in Distributed MoE Inference

ArXiv CS.AI **50%**

arXiv:2608.07964v1 Announce Type: cross Abstract: Load Balancing has emerged as a critical problem in expert-parallel distributed inference of Mixture-of-Experts (MoE) models. As routing distributions are typically skewed across experts, devices hosting lighter-loaded experts...

https://arxiv.org/abs/2608.07964

57 **CONF**

Verication-driven closed-loop multi-agent large language modelframework for code-compliant structural design

ArXiv CS.AI **50%**

arXiv:2608.07978v1 Announce Type: cross Abstract: Multi-agent large language model(LLM)systems are applied to structural design,yet most use one-shot generation and cannot verify their output,leaving themill-suited to safety-critical tasks.Rather than trusting LLM...

https://arxiv.org/abs/2608.07978

58 **CONF**

Do All LLMs Know When They're Being Harmful? A Reproducibility Study of Latent-Space Safety Probes Across Model Families

ArXiv CS.AI **50%**

arXiv:2608.08029v1 Announce Type: cross Abstract: Khatri et al. (2026) [DOI: 10.1109/DSN-W70714.2026.00027] show that lightweight MLP probes on final-layer activations of a single 8B model (LLaMA-3.1-8B) detect harmful prompts at F1 competitive with guard models 1000x larger,...

https://arxiv.org/abs/2608.08029

59 **CONF**

RAG-Audio: Retrieval-Augmented Generation for Faithful Brain-to-Audio Reconstruction

ArXiv CS.AI **50%**

arXiv:2608.09331v1 Announce Type: cross Abstract: Brain-to-audio reconstruction is limited by \emph{prior domination}: when a pretrained generator is conditioned on a weak neural signal, it produces realistic but stimulus-inaccurate audio. We introduce RAG-Audio, which decodes...

https://arxiv.org/abs/2608.09331

60 **CONF**

NeuPAT: Neuron-aware Plasticity Allocation Tuning for Language-Preserving MLLMs

ArXiv CS.AI **50%**

arXiv:2608.08107v1 Announce Type: cross Abstract: Multimodal expansion of large language models (LLMs) enables new perceptual capabilities but often compromises the language intelligence acquired during pretraining. In this work, we investigate this phenomenon from the...

https://arxiv.org/abs/2608.08107

61 **CONF**

Can LLM Agents Stick to the Script? A Benchmark for Long-Horizon Consistency in Interactive Narratives

ArXiv CS.AI 50%

arXiv:2608.08160v1 Announce Type: cross Abstract: The rapid advancement of Large Language Models (LLMs) is revolutionizing AI for Games by enabling open-ended and fluid interactive storytelling. However, existing research has largely overlooked the critical challenge of...

https://arxiv.org/abs/2608.08160

62 **CONF**

Privacy-Preserving Data Drift Detection and Recovery for Large-Scale LLM Applications via Proxy Representations

ArXiv CS.AI 50%

arXiv:2608.08245v1 Announce Type: cross Abstract: LLM applications deployed at scale face a fundamental challenge: privacy constraints prevent direct inspection of user interactions, making it difficult to obtain any representative evaluation dataset or to track the ongoing...

https://arxiv.org/abs/2608.08245

63 **CONF**

On the Robustness of LLMs' Internal Representation of Code Correctness

ArXiv CS.AI 50%

arXiv:2608.08266v1 Announce Type: cross Abstract: Code generated by modern language models often reads naturally. Yet, it also often fails to implement what was asked. This should be no surprise, as research shows the models' own confidence signals are poorly calibrated with...

https://arxiv.org/abs/2608.08266

64 **CONF**

Private Etymology: Designing Relational Reuse of Shared Symbols in Long-Term Human-AI Interaction

ArXiv CS.AI 50%

arXiv:2608.08443v1 Announce Type: cross Abstract: Previous studies have shown that people can develop shared symbols, partner-specific expressions, personal idioms, inside jokes, and other parts of a relational microculture. Recent work has also examined how humans and...

https://arxiv.org/abs/2608.08443

65 **CONF**

Calling the Bluff: Detecting Ever-Shifting Harmful Chat Dialogue via Ordered Reasoning Chain Regularization

ArXiv CS.AI 50%

arXiv:2608.08451v1 Announce Type: cross Abstract: Harmful chat dialogues are ever-shifting through type-shifting and lexical evasion, yet we find they share invariant principles, i.e., an Ordered Reasoning Chain (ORC) of recurring topics, harm language indicators, severity...

https://arxiv.org/abs/2608.08451

66 **CONF**

Beyond Tables: Doc2DB-Bench for Relationally Faithful Document-to-Database Construction

ArXiv CS.AI 50%

arXiv:2608.08459v1 Announce Type: cross Abstract: Practical AI systems increasingly need to turn long, heterogeneous documents into queryable relational databases, not isolated spreadsheets. In domains such as finance, healthcare, education, transportation, and enterprise...

https://arxiv.org/abs/2608.08459

67 **CONF**

On-Device Multi-Species Malaria Detection with Uncertainty-Calibrated Slide-Level Aggregation

ArXiv CS.AI **50%**

arXiv:2608.08566v1 Announce Type: cross Abstract: Malaria remains a leading cause of mortality in resource-limited settings, where expert microscopists are scarce. Automated diagnosis based on microscopy images thus has strong potential to improve care delivery. But for an...

https://arxiv.org/abs/2608.08566

68 **CONF**

RAG-Based Auto-Configuration for Industrial Fieldbus Devices

ArXiv CS.AI **50%**

arXiv:2608.08618v1 Announce Type: cross Abstract: Industrial device commissioning requires engineers to manually extract hundreds of protocol-specific parameters from heterogeneous PDF manuals and transcribe them into supervisory control systems, a time-intensive, error-prone...

https://arxiv.org/abs/2608.08618

69 **CONF**

LibraSpec: Dynamic Diffusion-Based Speculative Decoding via Marginal-Gain-Driven Optimization

ArXiv CS.AI **50%**

arXiv:2608.08721v1 Announce Type: cross Abstract: Speculative decoding accelerates large language model inference by drafting multiple tokens for parallel verification, with efficiency critically determined by the speculative length selected at each decoding round. Existing...

https://arxiv.org/abs/2608.08721

70 **CONF**

Can We Optimize the Performance-Carbon Emission Break-Even Point?: The Quest for Greener LLMs

ArXiv CS.AI **50%**

arXiv:2608.08744v1 Announce Type: cross Abstract: The carbon footprint of any deployed Large Language Model (LLM) accumulates during inference, where repeated use of the model substantially exceeds the one-time cost of fine-tuning. Yet most efficiency interventions target either...

https://arxiv.org/abs/2608.08744

71 **CONF**

360CityArena: A Realistic Virtual Urban Navigation Benchmark for Embodied Agents

ArXiv CS.AI **50%**

arXiv:2608.08814v1 Announce Type: cross Abstract: We present 360CityArena, a benchmark for evaluating the urban exploration capabilities of embodied agents within a photorealistic environment constructed from 360-degree videos. Existing outdoor benchmarks either lack sufficient...

https://arxiv.org/abs/2608.08814

72 **CONF**

Hybrid Neural-Classical Correction for Frozen Time Series Foundation Models: A Comprehensive Ablation Study on High-Frequency Stock Prediction

ArXiv CS.AI **50%**

arXiv:2608.08825v1 Announce Type: cross Abstract: Foundation models for time series forecasting demonstrate impressive zero-shot generalization but often underperform on specialized domains such as high-frequency finance. We present a comprehensive study of hybrid...

https://arxiv.org/abs/2608.08825

73 CONF

Agentic Anomaly Detection with ORCA-Style Dynamic Inductive Bias Adaptation in Multimodal Wearable Time Series Data

ArXiv CS.AI 50%

arXiv:2608.08859v1 Announce Type: cross Abstract: Wireless Body Area Networks (WBANs) generate multivariate physiological time series that are highly nonstationary and must often be processed under strict computational and memory constraints. A critical yet underexplored...

https://arxiv.org/abs/2608.08859

74 CONF

From Recovery to Drop-off: How Action Post-training Reduces a VLM's Late-Layer Depth Decodability

ArXiv CS.AI 50%

arXiv:2608.08904v1 Announce Type: cross Abstract: How much of a vision-language model's (VLM) spatial understanding remains after the action post-training process of building a vision-language-action model (VLA)? We probe depth perception, a primitive of spatiogeometric...

https://arxiv.org/abs/2608.08904

75 CONF

Toward CT-Equivalent Image Quality in Low-Dose Radiotherapy Planning: Conditional Diffusion-Based CBCT-to-CT Synthesis and the Impact of CBCT Input Representation

ArXiv CS.AI 50%

arXiv:2608.08919v1 Announce Type: cross Abstract: During standard radiotherapy planning, repeated CT acquisitions are often required for patient registration, verification, and adaptive planning, resulting in increased cumulative X-ray dose. To mitigate this, low-dose cone-beam...

https://arxiv.org/abs/2608.08919

76 CONF

From Operational Design Domain to Action: A Systematic Behavioral Taxonomy for Autonomous Driving

ArXiv CS.AI 50%

arXiv:2608.08941v1 Announce Type: cross Abstract: Operational Design Domain (ODD) specifications describe where an automated driving system (ADS) is permitted to operate, but they do not prescribe what the ADS must demonstrably do once deployed within that domain. This gap...

https://arxiv.org/abs/2608.08941

77 CONF

Do AI Forecast Ensembles Sample the Correct Conditional Distribution?

ArXiv CS.AI 50%

arXiv:2608.08954v1 Announce Type: cross Abstract: Ensemble forecasting aims to sample the conditional distribution of outcomes; whether AI forecast ensembles do this correctly in a joint sense remains largely untested. We train a diffusion model for probabilistic subseasonal...

https://arxiv.org/abs/2608.08954

78 CONF

Fourier Self-Supervision for Fine-Grained Generalized Category Discovery

ArXiv CS.AI 50%

arXiv:2608.08963v1 Announce Type: cross Abstract: Generalized Category Discovery aims to recognize known categories while identifying novel ones within unlabeled data. Existing methods, typically based on self-supervision and contrastive learning, often struggle to capture...

https://arxiv.org/abs/2608.08963

79 CONF

GALA: Graph-Augmented LLM Agents for Root Cause Analysis and Incident Response in Microservices

ArXiv CS.AI 50%

arXiv:2608.08968v1 Announce Type: cross Abstract: Microservice root cause analysis (RCA) requires correlating failures across heterogeneous telemetry within complex service dependency graphs. Existing methods often rely on a single telemetry modality; recent LLM-based approaches...

https://arxiv.org/abs/2608.08968

80 CONF

How Can Rhetoric Reward-Hack AI Reviewers? Dissecting Rhetorical Sensitivity in AI-Based Peer Review

ArXiv CS.AI 50%

arXiv:2608.08975v1 Announce Type: cross Abstract: As large language models increasingly participate in scientific evaluation, we investigate a potential form of reward hacking: how rhetorical choices shape AI-review judgments when reported scientific content is preserved and how...

https://arxiv.org/abs/2608.08975

81 CONF

How Far Do Foundation Models Transfer to Infant Signals? A Cross-Dataset Transfer Audit with a Unified Need Ontology

ArXiv CS.AI 50%

arXiv:2608.08989v1 Announce Type: cross Abstract: Public infant cry corpora are small, label-incompatible, and almost always evaluated one corpus at a time. We ask what this practice hides and what fixes it. Across four cry corpora screened by a multi-level leakage audit...

https://arxiv.org/abs/2608.08989

82 CONF

Guardian Crawler: Retrieval-First Knowledge Discovery with Bounded LLM Augmentation for Noisy Web Intelligence

ArXiv CS.AI 50%

arXiv:2608.08994v1 Announce Type: cross Abstract: Retrieving relevant evidence from noisy web data is challenging, particularly in sensitive domains containing incomplete reports, heterogeneous language, and irrelevant content. We present Guardian Crawler, a reproducible...

https://arxiv.org/abs/2608.08994

83 CONF

DeepFreqMark: End-To-End Learnable Frequency-Domain Watermarking with Spherical Attack Simulation for Latent Diffusion Models

ArXiv CS.AI 50%

arXiv:2608.08999v1 Announce Type: cross Abstract: The proliferation of AI-generated images produced by Latent Diffusion Models (LDMs) has raised critical concerns regarding copyright infringement and misinformation. Although existing frequency-domain watermarking methods embed...

https://arxiv.org/abs/2608.08999

84 CONF

SignLlama: Enhancing Gloss-free Sign Language Translation by Prioritizing Visual Features for LLMs

ArXiv CS.AI 50%

arXiv:2608.09006v1 Announce Type: cross Abstract: Large Language Models (LLMs) have achieved remarkable success across a wide range of tasks. However, fine-tuning LLMs for Gloss-Free Sign Language Translation (GFSLT) remains a challenge. In this paper, we investigate how to...

https://arxiv.org/abs/2608.09006

85 CONF

How People Evaluate AI-, Expert-, and Peer-Style Financial Advice

ArXiv CS.AI 50%

arXiv:2608.09019v1 Announce Type: cross Abstract: As generative AI increasingly becomes a common source of daily decision-making, including financial choices, it is critical to understand how people evaluate AI-generated financial advice. We conducted a preregistered vignette...

https://arxiv.org/abs/2608.09019

86 CONF

Triple Expert Learning from Noisy Labels for Semi-Supervised Vision Foundation Model Adaptation

ArXiv CS.AI 50%

arXiv:2608.09052v1 Announce Type: cross Abstract: Semi-supervised adaptation of vision foundation models (VFMs) commonly freezes the pretrained backbone and updates lightweight modules such as LoRA. However, pseudo-labels have mixed reliability, and a single LoRA adapter must...

https://arxiv.org/abs/2608.09052

87 CONF

A Unified Issue Resolution Benchmark for Requirement Clarification, Planning, and Code Generation for Coding Agents

ArXiv CS.AI 50%

arXiv:2608.09072v1 Announce Type: cross Abstract: Large language model-powered coding agents are increasingly used to modify existing code repositories, for example, by adding features or fixing bugs. Yet existing repository-level benchmarks typically evaluate only whether the...

https://arxiv.org/abs/2608.09072

88 CONF

When Confidence Fails: Overconfidence in LLMs under Uncertainty and Missing Clinical Information

ArXiv CS.AI 50%

arXiv:2608.09080v1 Announce Type: cross Abstract: Large Language Models (LLMs) have achieved strong performance in medical question answering and clinical reasoning tasks. However, their reliability under uncertainty remains poorly understood which raises critical concerns for...

https://arxiv.org/abs/2608.09080

89 CONF

The Announcement Carries the Cue: Markup, Boundaries, and the Notation of Pre-Training Corpora

ArXiv CS.AI 50%

arXiv:2608.09093v1 Announce Type: cross Abstract: How a document's arrangement is written down, its notation, is a training variable that no dataset card records. The field has established that text-extraction choices change model behaviour, and has never once measured the...

https://arxiv.org/abs/2608.09093

90 CONF

Visual Distortion Detection in UGC Images Using Large Multimodal Models

ArXiv CS.AI 50%

arXiv:2608.09122v1 Announce Type: cross Abstract: The localized depiction of perceptual quality has long been a crucial, yet underexplored, challenge in image quality assessment (IQA). Existing approaches based on large multimodal models (LMMs) predominantly rely on text-driven...

https://arxiv.org/abs/2608.09122

91 CONF

From Inaudible Inputs to Model Failures: Low-Frequency Safety Risks in LALMs

ArXiv CS.AI 50%

arXiv:2608.09158v1 Announce Type: cross Abstract: Large audio-language models (LALMs) have demonstrated strong capabilities in understanding diverse audio inputs. This diversity includes low-frequency signals that are inaudible to humans but can still enter the model and...

https://arxiv.org/abs/2608.09158

92 CONF

AkasicDB: Demonstrating Omni RAG with a Unified Vector-Graph-Relational DBMS

ArXiv CS.AI 50%

arXiv:2608.09214v1 Announce Type: cross Abstract: Recent Retrieval-Augmented Generation (RAG) systems increasingly combine vector retrieval with structured knowledge, such as Graph RAG and Filtered vector search. However, existing database architectures struggle to support such...

https://arxiv.org/abs/2608.09214

93 CONF

Beyond Solvability: Task Learnability as a Static Prior for LLM RL Post-Training

ArXiv CS.AI 50%

arXiv:2608.09217v1 Announce Type: cross Abstract: Reinforcement learning (RL) has become a central post-training paradigm for eliciting reasoning capabilities in large language models, yet uniform task sampling allocates compute without regard to differences in how tasks respond...

https://arxiv.org/abs/2608.09217

94 CONF

Governing the KV Cache: Preventing Timing Side-Channel Leakage in Multi-Tenant LLM Inference

ArXiv CS.AI 50%

arXiv:2608.09225v1 Announce Type: cross Abstract: The key-value (KV) cache is the primary throughput optimization in modern large language model (LLM) inference, enabling prefix reuse across requests. In multi-tenant deployments this cache is shared across tenants, creating a...

https://arxiv.org/abs/2608.09225

95 CONF

SafeQL: Search-based Refinement for Safe and Efficient LLM-based Text-to-SQL

ArXiv CS.AI 50%

arXiv:2608.09260v1 Announce Type: cross Abstract: Large language models (LLMs) have advanced Text-to-SQL by enabling natural language interfaces to databases without task-specific fine-tuning. However, existing LLM-based systems remain unreliable, often generating SQL queries...

https://arxiv.org/abs/2608.09260

96 CONF

GRASP: Granularity-Aware Region Alignment and Semantic Prototype Learning for Fine-Grained Cross-Modal Understanding in Drone Views

ArXiv CS.AI 50%

arXiv:2608.09270v1 Announce Type: cross Abstract: Fine-grained cross-modal understanding in drone views is essential for aerial vision-language navigation. However, the inherent wide field of view and overhead perspective of drone scenarios impose dual challenges on...

https://arxiv.org/abs/2608.09270

97 **CONF**

GLocFM: A Geometry-Aware Foundation Model for 3D Indoor Wireless Localization

ArXiv CS.AI 50%

arXiv:2608.09285v1 Announce Type: cross Abstract: Learning-based wireless localizers often fail to utilize geometric information about the propagation environment, limiting their ability to exploit non-line-of-sight (NLoS) propagation and generalize across scenes. To bridge this...

https://arxiv.org/abs/2608.09285

98 **CONF**

WorldSimProbe: Diagnosing Simulator Faithfulness in Action-Conditioned World Models for Embodied Manipulation

ArXiv CS.AI 50%

arXiv:2608.09298v1 Announce Type: cross Abstract: Action-conditioned world models (ACWMs) promise to provide embodied AI with scalable predictive simulators for planning, policy evaluation, and data generation. Realizing this promise requires precise action-conditioned...

https://arxiv.org/abs/2608.09298

99 **CONF**

Test-Time Augmentation for LLMs: When Input Diversity Beats Output Diversity at Matched Compute

ArXiv CS.AI 50%

arXiv:2608.09351v1 Announce Type: cross Abstract: Test-time scaling improves LLM accuracy but multiplies inference cost, making the accuracy gained per unit of compute the metric that matters in deployment. Self-consistency is one of the established approaches, which spends this...

https://arxiv.org/abs/2608.09351

00 **CONF**

Deep Learning based Detection of Fishing Vessels and Fishing Monitoring using Nightlight Images

ArXiv CS.AI 50%

arXiv:2608.09360v1 Announce Type: cross Abstract: The demand for maritime surveillance has given rise to the need for monitoring fishing vessel activities, particularly in addressing the challenge of "dark vessels" that operate without Automatic Identification System (AIS)...

https://arxiv.org/abs/2608.09360

01 **CONF**

Imaginative Generative AI: Crossing the Entropy Wall into Worlds Beyond Imitation

ArXiv CS.AI 50%

arXiv:2608.09385v1 Announce Type: cross Abstract: Generative AI models are primarily designed to imitate the data distribution, an objective that neither corrects diversity lost by a learned generator nor defines how generation should extend beyond the diversity of the data...

https://arxiv.org/abs/2608.09385

02 **CONF**

Continual Visual Anomaly Detection on the Edge: Benchmark and Efficient Solutions

ArXiv CS.AI 50%

arXiv:2604.06435v2 Announce Type: replace-cross Abstract: Visual Anomaly Detection (VAD) is a critical task for many applications including industrial inspection and healthcare. While VAD has been extensively studied, two key challenges remain largely unaddressed in conjunction:...

https://arxiv.org/abs/2604.06435

03 **CONF**

RecoverFly: A Failure-Aware Reinforcement Learning Post-Training Framework for Aerial Vision-Language Navigation

ArXiv CS.AI 50%

arXiv:2608.09467v1 Announce Type: cross Abstract: Unmanned aerial vehicle vision-language navigation (UAV-VLN) requires agents to translate visual observations and language instructions into reliable flight actions in complex environments. Although recent end-to-end UAV...

https://arxiv.org/abs/2608.09467

04 **CONF**

MixFormer: Linear Transformer with Mixture of Memory Experts

ArXiv CS.AI 50%

arXiv:2608.09468v1 Announce Type: cross Abstract: State Space Models (SSMs), as a mainstream research direction of linear Transformers, aim to achieve higher efficiency than standard Transformers in long-context modeling. However, existing SSMs suffer from limited input...

https://arxiv.org/abs/2608.09468

05 **CONF**

Beyond Uniform Restoration: Empowering All-in-One Restoration with Pixel-Level Multimodal Guidance

ArXiv CS.AI 50%

arXiv:2608.09482v1 Announce Type: cross Abstract: All-in-one image restoration is a unified low-level vision task that aims to effectively recover high-quality images from inputs degraded by various types and levels of corruption using a single model. Recent works have achieved...

https://arxiv.org/abs/2608.09482

06 **CONF**

Build it, Break it, Repeat: Benchmarking and improving LLM-manipulated disinformation detection in social media posts

ArXiv CS.AI 50%

arXiv:2608.09510v1 Announce Type: cross Abstract: Detecting machine-generated disinformation on social media is increasingly difficult as large language models (LLMs) make it easier to generate and rewrite misleading content at scale. Static benchmark evaluations, measuring...

https://arxiv.org/abs/2608.09510

07 **CONF**

STAIR: Effective Incident Response Using an End-to-End Agentic Planning Framework

ArXiv CS.AI 50%

arXiv:2608.09524v1 Announce Type: cross Abstract: Incident response planning is critical for restoring compromised software systems after cyberattacks. Common practice relies on expert-driven playbooks that encode fixed response procedures, but these static workflows struggle to...

https://arxiv.org/abs/2608.09524

08 **CONF**

NormAct: Benchmarking Embodied Agents' Proactive Compliance with Unspoken Social Norms

ArXiv CS.AI 50%

arXiv:2606.27826v3 Announce Type: replace Abstract: Embodied agents driven by multimodal large language models (MLLMs) can often complete everyday tasks from visual observations, but goal achievement does not establish whether they proactively respect unstated social norms....

https://arxiv.org/abs/2606.27826

09 CONF

Dual-Adversarial Safety Alignment: Cultivating Intrinsic Threat Comprehension in LRMs

ArXiv CS.AI 50%

arXiv:2608.09542v1 Announce Type: cross Abstract: Large reasoning models (LRMs) achieve remarkable success on complex tasks but remain vulnerable to harmful prompts that induce unsafe outputs. Recent methods align LRMs using direct refusals or safety rationales, yet often focus...

https://arxiv.org/abs/2608.09542

10 CONF

LEED: Local Embedding Evolution Distance for over-smoothing estimation and virtual node selection in GNN

ArXiv CS.AI 50%

arXiv:2608.09596v1 Announce Type: cross Abstract: Graph Neural Networks (GNNs) suffer from two fundamental limitations: over-smoothing, where node representations become indistinguishable with depth, and over-squashing, where long-range information is compressed through limited...

https://arxiv.org/abs/2608.09596

11 CONF

Measuring the Wrong Thing: Internal Harmfulness Scores Anti-Rank Successful Jailbreaks

ArXiv CS.AI 50%

arXiv:2608.09624v1 Announce Type: cross Abstract: Internal safety scores judge a prompt before any text is generated, and they are validated by how well they separate harmful prompts from benign ones. That separation is then read as evidence that the score will also catch the...

https://arxiv.org/abs/2608.09624

12 CONF

How Do Large Language Models Judge Social Attraction? Evidence from Theory-Grounded Persona Ratings Across Multiple LLMs and Humans

ArXiv CS.AI 50%

arXiv:2608.09717v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly used to perform subjective evaluations traditionally made by humans, yet their validity as social judges remains unclear. This paper examines whether LLMs can assess social attraction...

https://arxiv.org/abs/2608.09717

13 CONF

KGCaRe: Explainable Complex Conditional Question Answering using Automatic Knowledge Graph Construction and Context Retrieval with LLMs

ArXiv CS.AI 50%

arXiv:2608.09779v1 Announce Type: cross Abstract: Answering complex conditional questions using Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) remains a challenge, particularly in domain-specific contexts where general-purpose LLMs and RAG tend to...

https://arxiv.org/abs/2608.09779

14 CONF

Stealing Reasoning Traces from Proprietary LLM APIs

ArXiv CS.AI 50%

arXiv:2608.09867v1 Announce Type: cross Abstract: Leading large language model providers now conceal their models' step-by-step reasoning, or chain-of-thought, to protect intellectual property and limit information leakage. Rather than storing these traces server-side, providers...

https://arxiv.org/abs/2608.09867

15 CONF

LEGO-Puzzles: How Good Are MLLMs at Multi-Step Spatial Reasoning?

ArXiv CS.AI 50%

arXiv:2503.19990v4 Announce Type: replace Abstract: Many real-world applications of spatial intelligence, such as robotic control, autonomous driving, and automated assembly, require spatial reasoning across multiple sequential steps. However, the extent to which current...

https://arxiv.org/abs/2503.19990

16 CONF

How Much Backtracking is Enough? Exploring the Interplay of SFT and RL in Enhancing LLM Reasoning

ArXiv CS.AI 50%

arXiv:2505.24273v2 Announce Type: replace Abstract: Recent advancements in large language models (LLMs) suggest that reinforcement learning (RL) effectively internalizes search strategies, yielding significant improvements on challenging reasoning tasks through extended chains...

https://arxiv.org/abs/2505.24273

17 CONF

EgoBrain: Synergizing Minds and Eyes For Human Action Understanding

ArXiv CS.AI 50%

arXiv:2506.01353v3 Announce Type: replace Abstract: The integration of brain-computer interfaces (BCIs), in particular electroencephalography (EEG), with artificial intelligence (AI) has shown tremendous promise in decoding human cognition and behavior from neural signals. In...

https://arxiv.org/abs/2506.01353

18 CONF

Reflex First, Reflect Later: Latency-Aware Embodied LLM Agents for Dynamic Response

ArXiv CS.AI 50%

arXiv:2506.07223v2 Announce Type: replace Abstract: Large language models (LLMs) have substantially improved the planning capabilities of embodied agents, enabling their deployment in dynamic and safety-critical environments. However, these settings expose a critical limitation:...

https://arxiv.org/abs/2506.07223

19 CONF

Med-CRAFT: An Information System for Explainable and Configurable Construction of Multimodal Medical QA Datasets

ArXiv CS.AI 50%

arXiv:2512.01045v2 Announce Type: replace Abstract: Data-intensive artificial intelligence applications increasingly rely on large-scale, high-quality, explainable, and reproducible datasets, yet the construction of such datasets often remains labor-intensive, weakly traceable,...

https://arxiv.org/abs/2512.01045

20 CONF

Agentic AI for Clustering, Relationship Discovery, and Semantic Trading in Prediction Markets

ArXiv CS.AI 50%

arXiv:2512.02436v2 Announce Type: replace Abstract: Prediction markets allow users to trade on outcomes of real-world events, but are prone to fragmentation with overlapping questions, implicit equivalences, and hidden contradictions across markets. We present an agentic AI...

https://arxiv.org/abs/2512.02436

21 **CONF**

Time-Series Forecasting in Safety-Critical Environments: An Open-Source Package for EU-AI-Act-Compliant Development / Zeitreihenprognose in sicherheitskritischen Umgebungen: Ein Open-Source-Paket f\"ur die KI-VO-konforme Entwicklung

ArXiv CS.AI 50%

arXiv:2604.23859v2 Announce Type: replace Abstract: With spotforecast2-safe we present an integrated Compliance-by-Design approach to Python-based point forecasting of time series in safety-critical environments. A review of the relevant open-source tooling shows that existing...

https://arxiv.org/abs/2604.23859

22 **CONF**

Human agency in initial human-AI proof formalization workflows

ArXiv CS.AI 50%

arXiv:2606.04273v2 Announce Type: replace Abstract: For centuries, human mathematicians have written proofs to substantiate their mathematical arguments; yet, the ability to automatically verify the validity of proofs has long been a challenge. Advances in AI systems' ability to...

https://arxiv.org/abs/2606.04273

23 **CONF**

SearchSwarm: Towards Delegation Intelligence in Agentic LLMs for Long-Horizon Deep Research

ArXiv CS.AI 50%

arXiv:2606.09730v2 Announce Type: replace Abstract: Large language models are increasingly expected to handle complex, long-horizon real-world tasks whose context demands can grow without bound, yet model context windows remain inherently finite. Recent work explores a paradigm...

https://arxiv.org/abs/2606.09730

24 **CONF**

RareLens: Towards End-to-End Rare Disease Care via Aligning Divergent Large Language Model Reasoning

ArXiv CS.AI 50%

arXiv:2607.23290v2 Announce Type: replace Abstract: Rare diseases represent one of the most challenging settings for clinical decision-making, where heterogeneous presentations, sparse evidence and limited expertise create persistent uncertainty throughout the care pathway....

https://arxiv.org/abs/2607.23290

25 **CONF**

InfoOps Bench: A live information operations safety benchmark

ArXiv CS.AI 50%

arXiv:2607.28503v3 Announce Type: replace Abstract: In this paper we present an active, constantly updated AI benchmark which measures the integrity of frontier language models against being co-opted for use by authoritarian state "information operations": intentional,...

https://arxiv.org/abs/2607.28503

26 **CONF**

Learning to Coordinate Symbolic Tools: LLM Agents for Verified Sum-of-Squares Certificates

ArXiv CS.AI 50%

arXiv:2608.00326v2 Announce Type: replace Abstract: Tool calling allows large language models (LLMs) to invoke external computation during problem solving, a useful capability in various fields including AI for mathematics. We study this setting through weighted sum-of-squares...

https://arxiv.org/abs/2608.00326

27 **CONF**

The Transformer Revolution, Part 1: Dynamic Processing through Output-Weight Interconnections

ArXiv CS.AI 50%

arXiv:2608.03921v2 Announce Type: replace Abstract: This paper offers a new interpretation of the Transformer during inference. Against the "stochastic parrot" view that large language models merely reproduce statistical regularities learned in training, we argue that...

https://arxiv.org/abs/2608.03921

28 **CONF**

ViSR-KGC: Visual Subgraph Reasoning with Vision-Language Models for Multimodal Knowledge Graph Completion

ArXiv CS.AI 50%

arXiv:2608.05833v2 Announce Type: replace Abstract: Knowledge graph completion (KGC) aims to infer missing entities or relations from incomplete graph structures, and has evolved into multimodal knowledge graph completion (MMKGC), where entities are associated with multiple...

https://arxiv.org/abs/2608.05833

29 **CONF**

Explainable Machine Learning-Based Security and Privacy Protection Framework for Internet of Medical Things Systems

ArXiv CS.AI 50%

arXiv:2403.09752v4 Announce Type: replace-cross Abstract: The Internet of Medical Things transcends traditional medical boundaries, enabling a transition from reactive treatment to proactive prevention. This innovative method revolutionizes healthcare by facilitating early...

https://arxiv.org/abs/2403.09752

30 **CONF**

LF^{2}AR: Accounting for Layerwise Dynamics to Improve Multimodal Adaptation of Language Models

ArXiv CS.AI 50%

arXiv:2503.06211v3 Announce Type: replace-cross Abstract: Text-pretrained language models (LMs) encode rich world knowledge, but adapting them to process and generate perceptual modalities such as audio and images while effectively leveraging that knowledge remains challenging....

https://arxiv.org/abs/2503.06211

31 **CONF**

TreeHop: Efficient Embedding-Level Query Rewriter

ArXiv CS.AI 50%

arXiv:2504.20114v3 Announce Type: replace-cross Abstract: Retrieval-augmented generation (RAG) systems face significant challenges in multi-hop question answering (MHQA), where complex queries require synthesizing information across multiple document chunks. Existing approaches...

https://arxiv.org/abs/2504.20114

32 **CONF**

Transformer-Based Neural Quantum Digital Twins for Many-Body Spectral Reconstruction and Adaptive Quantum-Annealing Schedule Design

ArXiv CS.AI 50%

arXiv:2505.15662v3 Announce Type: replace-cross Abstract: We introduce Transformer-based Neural Quantum Digital Twins (Tx-NQDTs) to reconstruct the low-energy spectral evolution of many-body quantum systems along quantum-annealing paths, including ground- and first-excited-state...

https://arxiv.org/abs/2505.15662

33 **CONF**

Transformer Circuits Can Realize Clustering Algorithms

ArXiv CS.AI **50%**

arXiv:2506.19125v2 Announce Type: replace-cross Abstract: Although transformers are most commonly optimized as statistical sequence models, it is unclear to what extent they can implement and learn exact algorithmic computations. Here, we specify a transformer implementation...

https://arxiv.org/abs/2506.19125

34 **CONF**

Dynamic gain neuromodulation attenuates the stability gap under joint training

ArXiv CS.AI **50%**

arXiv:2507.14056v3 Announce Type: replace-cross Abstract: Recent work in continual learning has highlighted the stability gap -- a temporary performance drop on previously learned tasks when new ones are introduced. This phenomenon reflects a mismatch between rapid adaptation...

https://arxiv.org/abs/2507.14056

35 **CONF**

Enhancing Knowledge Tracing through Leakage-Free and Recency-Aware Embeddings

ArXiv CS.AI **50%**

arXiv:2508.17092v2 Announce Type: replace-cross Abstract: Knowledge Tracing (KT) aims to predict a student's future performance based on their sequence of interactions with learning content. Many KT models rely on knowledge concepts (KCs), which represent the skills required for...

https://arxiv.org/abs/2508.17092

36 **CONF**

Deep Residual Echo State Networks: exploring residual orthogonal connections in untrained Recurrent Neural Networks

ArXiv CS.AI **50%**

arXiv:2508.21172v3 Announce Type: replace-cross Abstract: Echo State Networks (ESNs) are a particular type of untrained Recurrent Neural Networks (RNNs) within the Reservoir Computing (RC) framework, popular for their fast and efficient learning. However, traditional ESNs often...

https://arxiv.org/abs/2508.21172

37 **CONF**

Topographic Constraints Shape Brain-Like Component Structure in Auditory Models

ArXiv CS.AI **50%**

arXiv:2509.24039v2 Announce Type: replace-cross Abstract: If topography is a fundamental feature of the brain, it should influence both how neurons are arranged in space (i.e. explain brain maps) and how information is structured within the neural population. The human auditory...

https://arxiv.org/abs/2509.24039

38 **CONF**

NeuroAda: Activating Each Neuron's Potential for Parameter-Efficient Fine-Tuning

ArXiv CS.AI **50%**

arXiv:2510.18940v2 Announce Type: replace-cross Abstract: Existing parameter-efficient fine-tuning (PEFT) methods primarily fall into two categories: addition-based and selective in-situ adaptation. The former, such as LoRA, introduce additional modules to adapt the model to...

https://arxiv.org/abs/2510.18940

39 CONF

Beyond Pixels: Benchmarking and Reward-Based Assessing Framework for Visual Spatial Aesthetics

ArXiv CS.AI 50%

arXiv:2512.05098v2 Announce Type: replace-cross Abstract: In recent years, Image Quality Assessment (IQA) for AI-generated images (AIGI) has advanced rapidly; however, existing methods primarily target portraits and artistic images, lacking a systematic evaluation of interior...

https://arxiv.org/abs/2512.05098

40 CONF

Bounding Hallucinations: Merlin-Arthur Protocols for Mutual-Information Bounds in Language Models

ArXiv CS.AI 50%

arXiv:2512.11614v3 Announce Type: replace-cross Abstract: Retrieval-augmented generation (RAG) relies on retrieved context to guide large language models (LLM), yet treats the retrieval as a heuristic rather than verifiable evidence -- leading to unsupported answers,...

https://arxiv.org/abs/2512.11614

41 CONF

LAUDE: LLM-Assisted Unit Test Generation and Debugging of Hardware DEsigns

ArXiv CS.AI 50%

arXiv:2601.08856v3 Announce Type: replace-cross Abstract: Unit tests are critical in the hardware design lifecycle to ensure that component design modules are functionally correct and conform to the specification before they are integrated at the system level. Thus developing...

https://arxiv.org/abs/2601.08856

42 CONF

The Cost of Language: Centroid Erasure Exposes and Exploits Modal Competition in Multimodal Language Models

ArXiv CS.AI 50%

arXiv:2604.14363v2 Announce Type: replace-cross Abstract: Multimodal language models systematically underperform on visual perception tasks, yet the structure underlying this failure remains poorly understood. We propose centroid replacement, mapping tokens to their nearest...

https://arxiv.org/abs/2604.14363

43 CONF

Culturally Situated AI Safety for Youth: Saudi Arabian Perspectives of Youth, Parents and Teachers

ArXiv CS.AI 50%

arXiv:2604.26494v2 Announce Type: replace-cross Abstract: Generative AI tools are widely used by youth and have introduced new privacy and safety challenges. While prior research has explored youths safety in GenAI within a Western context, it often overlooks the cultural,...

https://arxiv.org/abs/2604.26494

44 CONF

HEART: Exploiting Head Heterogeneity in Sparse Attention for Video Diffusion

ArXiv CS.AI 50%

arXiv:2605.14513v2 Announce Type: replace-cross Abstract: Sparse attention accelerates video diffusion by allowing each attention head to focus on only a small subset of interactions. Existing methods already construct head-specific sparse patterns conditioned on the input....

https://arxiv.org/abs/2605.14513

45 CONF

Toward Measuring AI's Effects on Skill Formation: The Stock-Formation Gap

ArXiv CS.AI 50%

arXiv:2605.16283v3 Announce Type: replace-cross Abstract: Large-scale AI deployment data and controlled learning experiments characterize different consequences of the same technology. Deployment telemetry shows that AI use is concentrated in skilled work and frequently supports...

https://arxiv.org/abs/2605.16283

46 CONF

Prompts Don't Protect: Architectural Enforcement via MCP Proxy for LLM Tool Access Control

ArXiv CS.AI 50%

arXiv:2605.18414v2 Announce Type: replace-cross Abstract: Large language models increasingly operate as autonomous agents that select and invoke tools from large registries. We identify a critical gap: when unauthorized tools are visible in an agent's context, models select them...

https://arxiv.org/abs/2605.18414

47 CONF

Don't Retrain, Just Reuse: Recovering Dual-Target Molecules from Single-Target Diffusion Models

ArXiv CS.AI 50%

arXiv:2605.25681v2 Announce Type: replace-cross Abstract: Designing a single molecule that modulates two targets is a promising strategy for polypharmacology, but it remains substantially harder than standard single-target generation because one candidate must satisfy two...

https://arxiv.org/abs/2605.25681

48 CONF

Beyond Questions: Evaluating LLM's Knowledge Expression

ArXiv CS.AI 50%

arXiv:2605.26937v2 Announce Type: replace-cross Abstract: Parametric knowledge in large language models (LLMs) is a cornerstone of their success, yet remains poorly understood. Existing knowledge benchmarks typically rely on predefined questions (e.g., "What is the birth date of...

https://arxiv.org/abs/2605.26937

49 CONF

Enhancing AI Interpretability with Localised Architectures

ArXiv CS.AI 50%

arXiv:2606.07998v3 Announce Type: replace-cross Abstract: Recent advances in generative AI, especially powerful Large Language Models (LLMs), raise concerns over the interpretability, safety and sustainability of these large and opaque AI models. The power of such architectures...

https://arxiv.org/abs/2606.07998

50 CONF

Decodable but Not Faithful: Coupling Natural-Language Rationales to Programmatic Verifiers

ArXiv CS.AI 50%

arXiv:2606.21678v2 Announce Type: replace-cross Abstract: Language models can generate plausible rationales for their predictions, but these explanations may not faithfully represent the model's internal reasoning. We propose verifier-coupled reasoning, a framework that inserts...

https://arxiv.org/abs/2606.21678

51 **CONF**

Scaling Audio Models Efficiently: A Joint Study of Compute Constraints and Optimization Behavior

ArXiv CS.AI 50%

arXiv:2606.22790v2 Announce Type: replace-cross Abstract: In this paper, we investigate the tradeoffs between compute allocation and model performance for two speech processing tasks: Automatic Speech Recognition (ASR) and Speech Emotion Recognition (SER). We propose a unified...

https://arxiv.org/abs/2606.22790

52 **CONF**

ATMA: Long-Context Language Modeling via Polar Attention and Gated-Delta Compression Memory

ArXiv CS.AI 50%

arXiv:2606.25156v3 Announce Type: replace-cross Abstract: Native length extrapolation remain a weakly solvable problem in language modeling due to trade-off balancing between exact retrieval fidelity, long-document likelihood, and inference efficiency. We present ATMA as a...

https://arxiv.org/abs/2606.25156

53 **CONF**

Safeguarding LLM Agents from Misalignment through Provenance Analysis

ArXiv CS.AI 50%

arXiv:2607.01236v2 Announce Type: replace-cross Abstract: As LLM agents gain increasing access to powerful tools, ensuring that their actions align with the user's intent becomes critical. When an agent's proposed action deviates from that intent---a phenomenon called...

https://arxiv.org/abs/2607.01236

54 **CONF**

Decoy Images Amplify Caption-Mediated Defenses Against Encoded Jailbreaks

ArXiv CS.AI 50%

arXiv:2608.01043v2 Announce Type: replace-cross Abstract: We report a counter-intuitive interaction between image inputs and existing black-box defenses on Vision--Language Models (VLMs): pairing an encoded jailbreak prompt with an unrelated decoy image can sharply lower attack...

https://arxiv.org/abs/2608.01043

55 **CONF**

Symbolic Attack Chain Generation from Atomic Red Team Techniques: An Empirical Study of Predicate Representation Granularity

ArXiv CS.AI 50%

arXiv:2608.00143v2 Announce Type: replace-cross Abstract: Automated attack chain generation is critical for modern cybersecurity, yet manual construction fails to scale as adversary behaviors expand. While classical AI planning using the Planning Domain Definition Language...

https://arxiv.org/abs/2608.00143